

ML Hybrid Models Comparison Report

Generated: 2025-11-24T22:36:46.462695

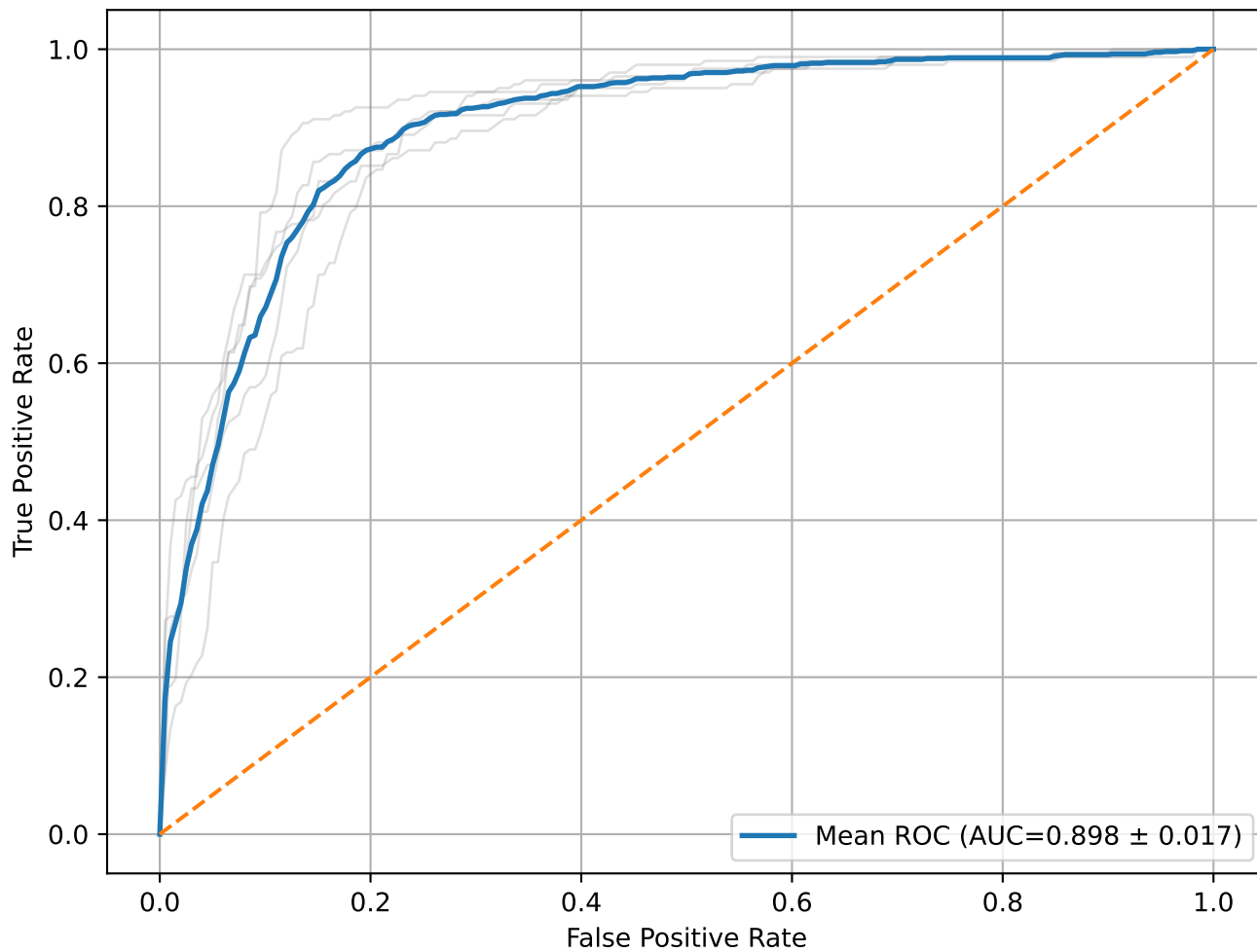
Dataset samples: 2009, features: 447

Models: logistic, rf, svm, xgb, lgb, dnn, stacking, voting, bagging, pca_svm, ga_xgb, pso_svm, bayes_opt_svms, gbd

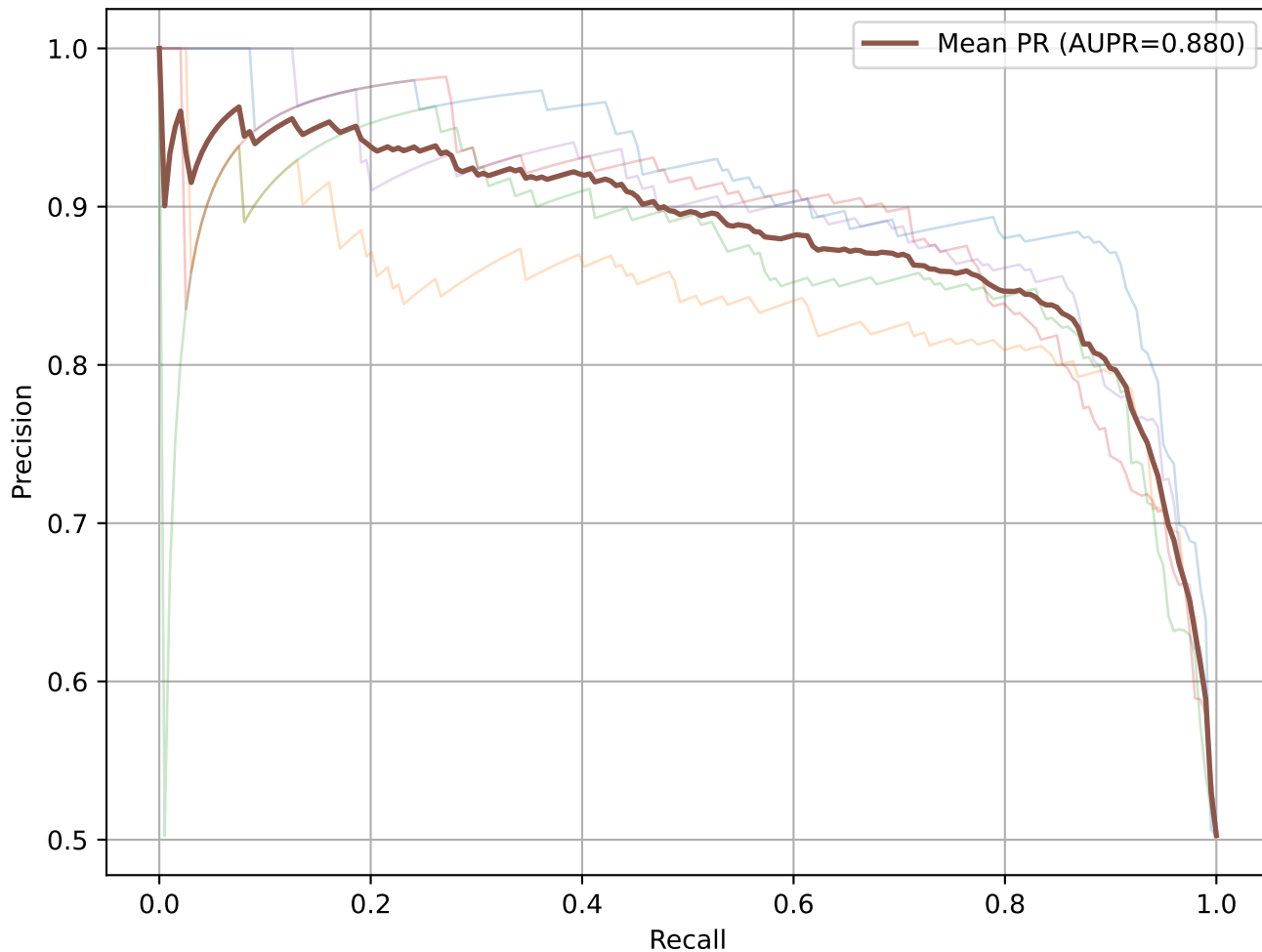
StratifiedKFold: 5 splits

Random seed: 42

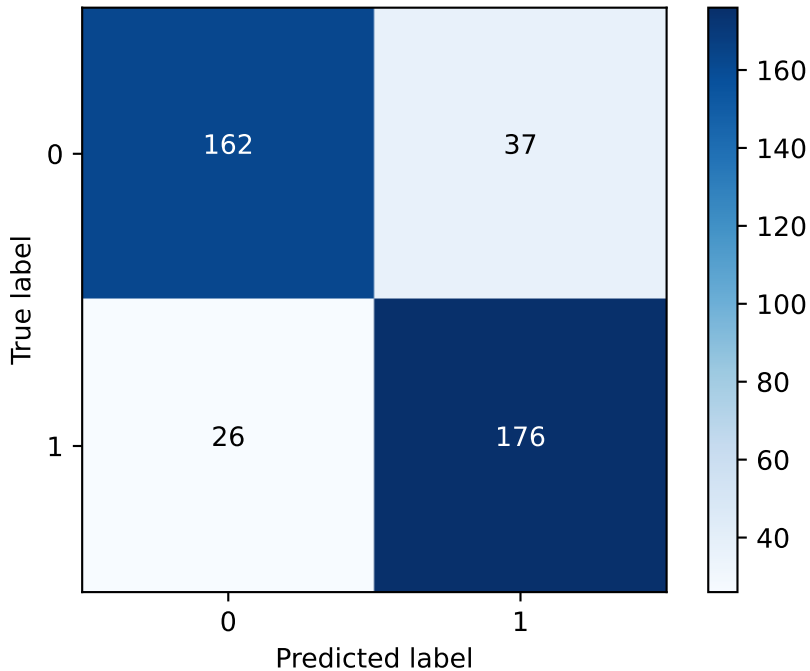
ROC - LOGISTIC



Precision-Recall - LOGISTIC



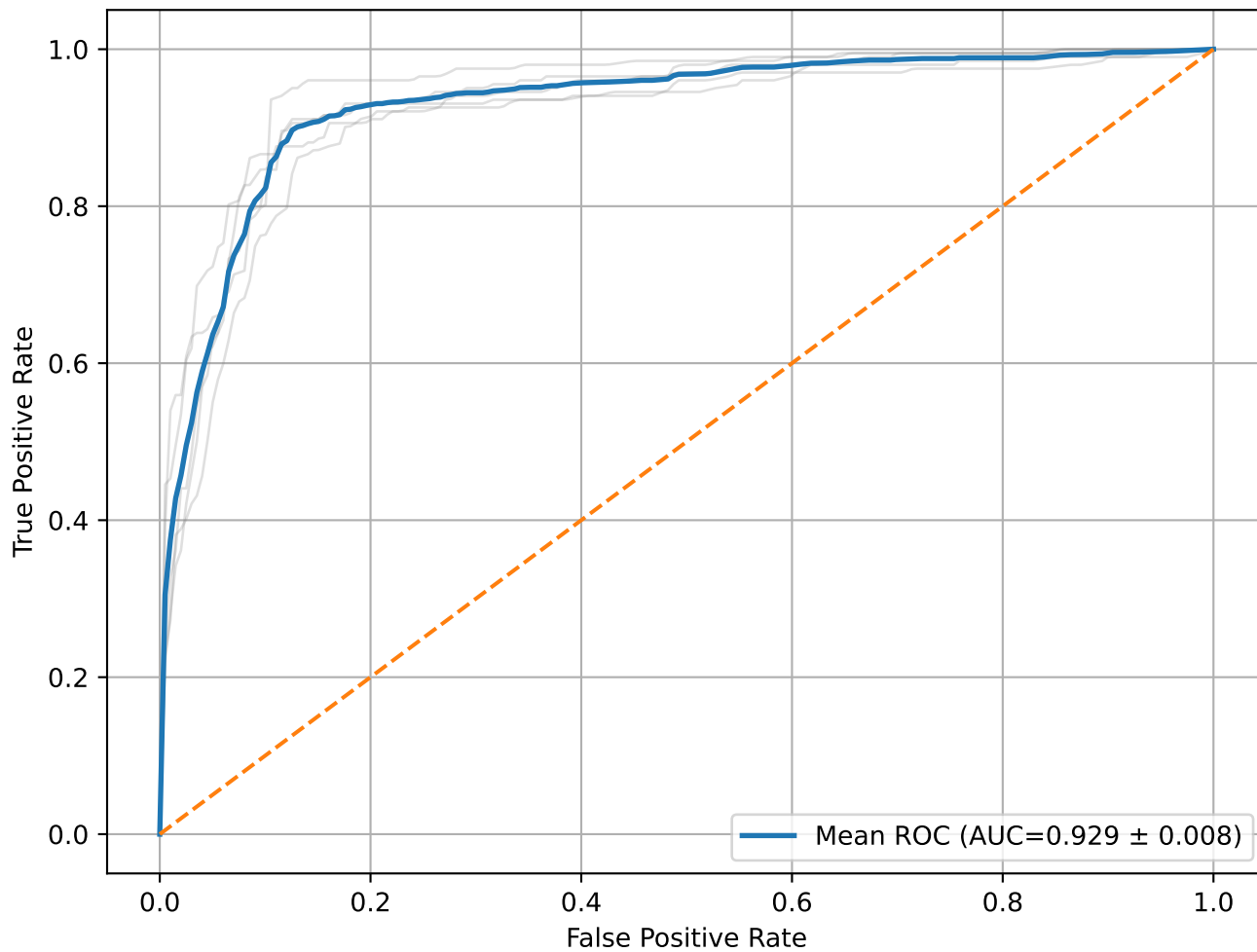
Confusion Matrix - LOGISTIC (last fold)



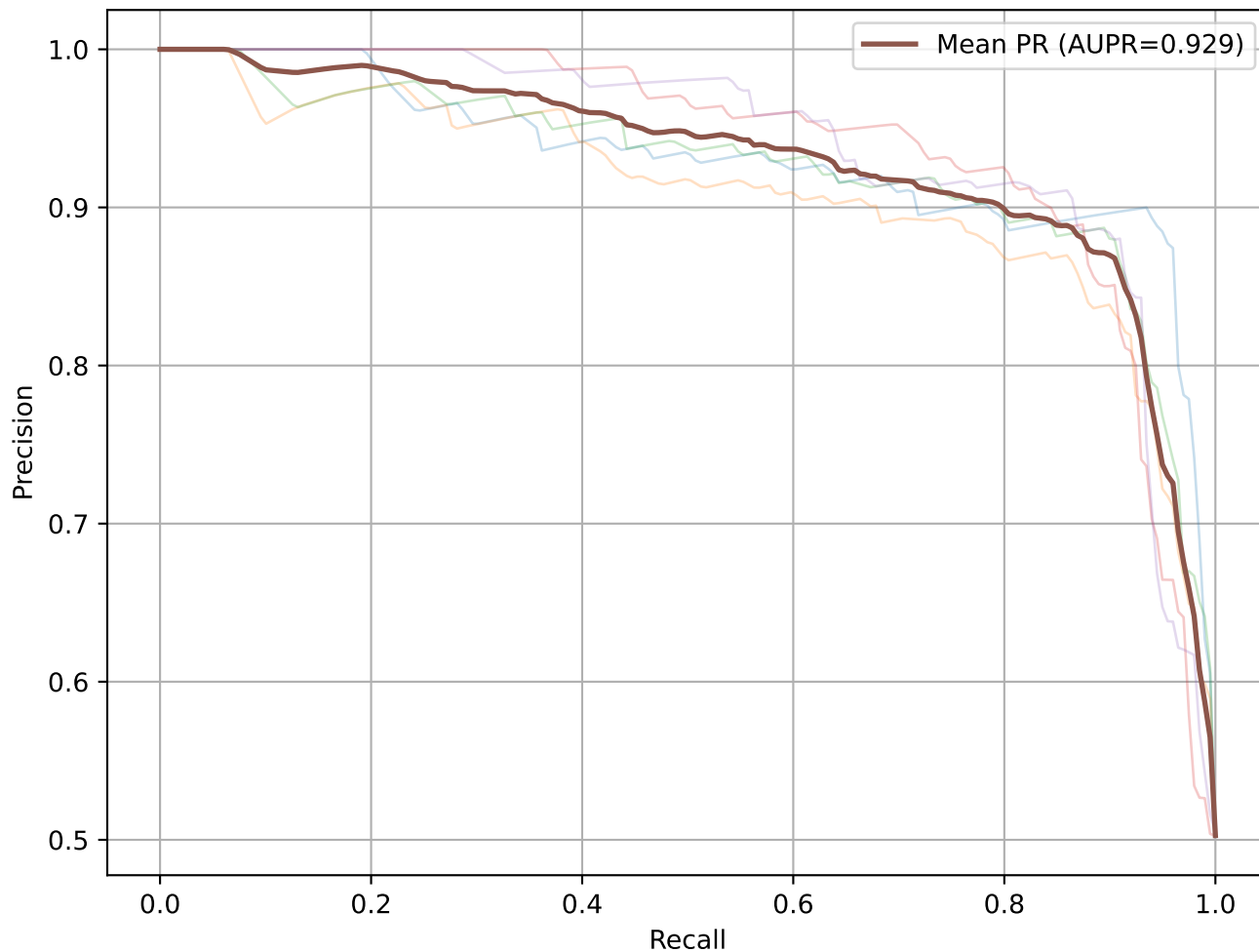
ClassificationReport - LOGISTIC (last fold)

	precision	recall	f1-score	support
0	0.86	0.81	0.84	199
1	0.83	0.87	0.85	202
accuracy			0.84	401
macro avg	0.84	0.84	0.84	401
weighted avg	0.84	0.84	0.84	401

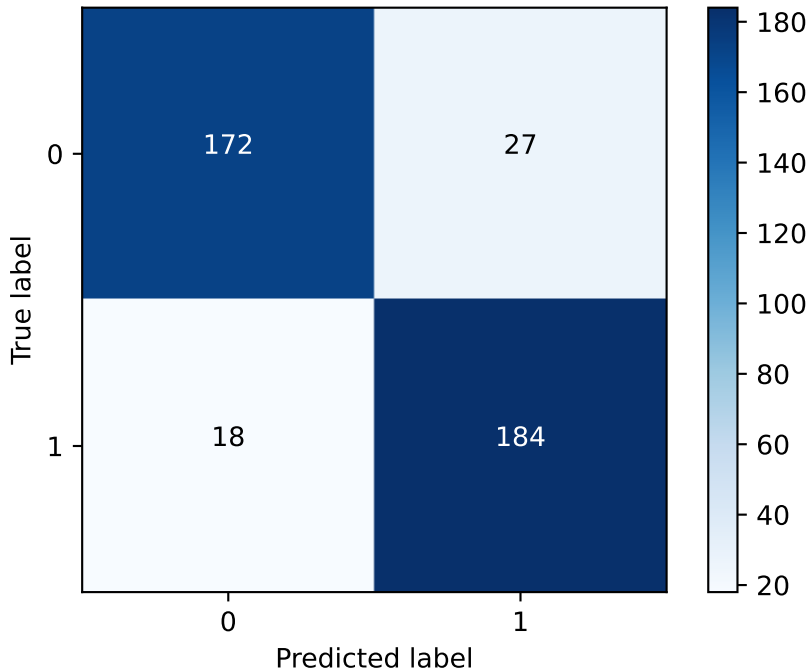
ROC - RF



Precision-Recall - RF



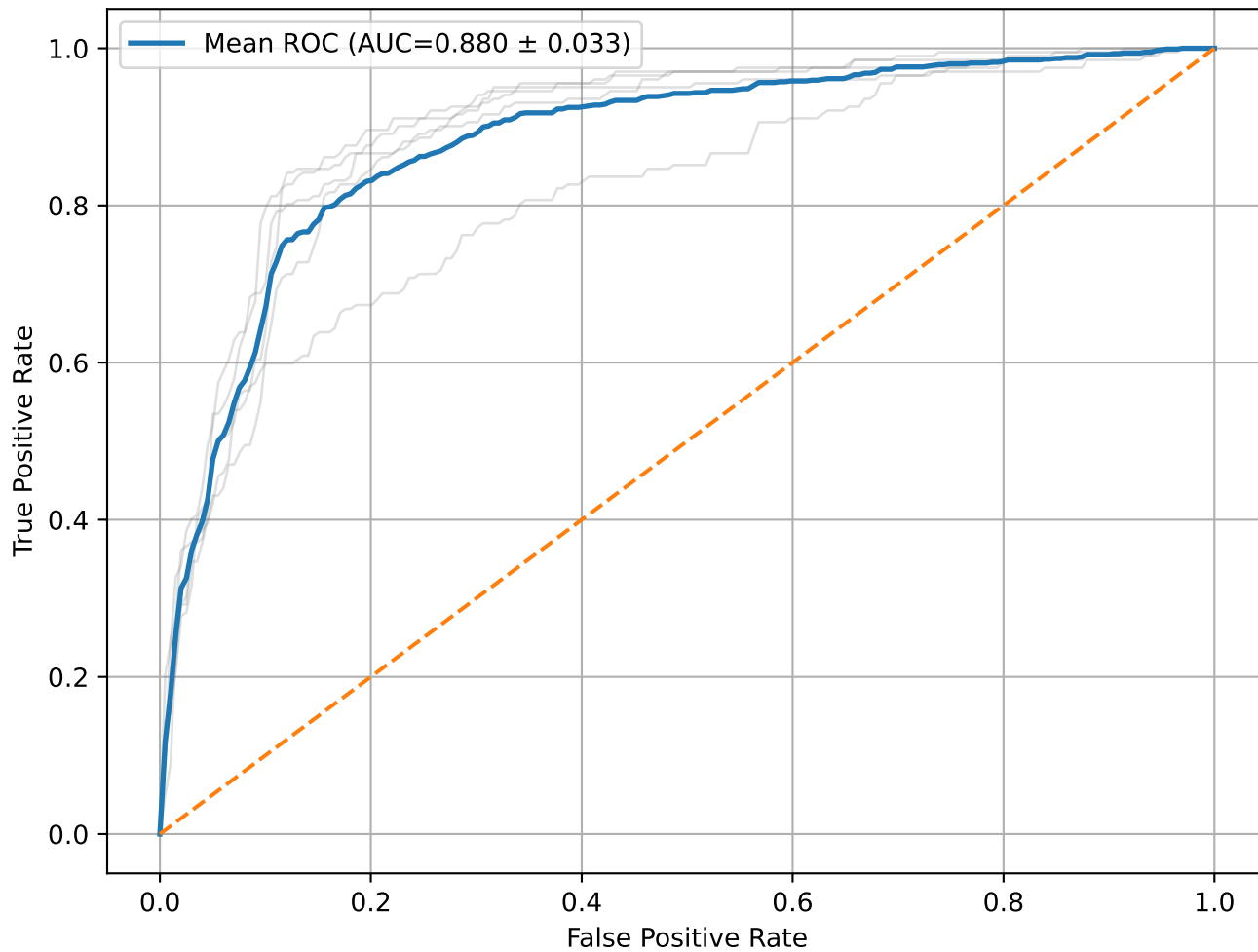
Confusion Matrix - RF (last fold)



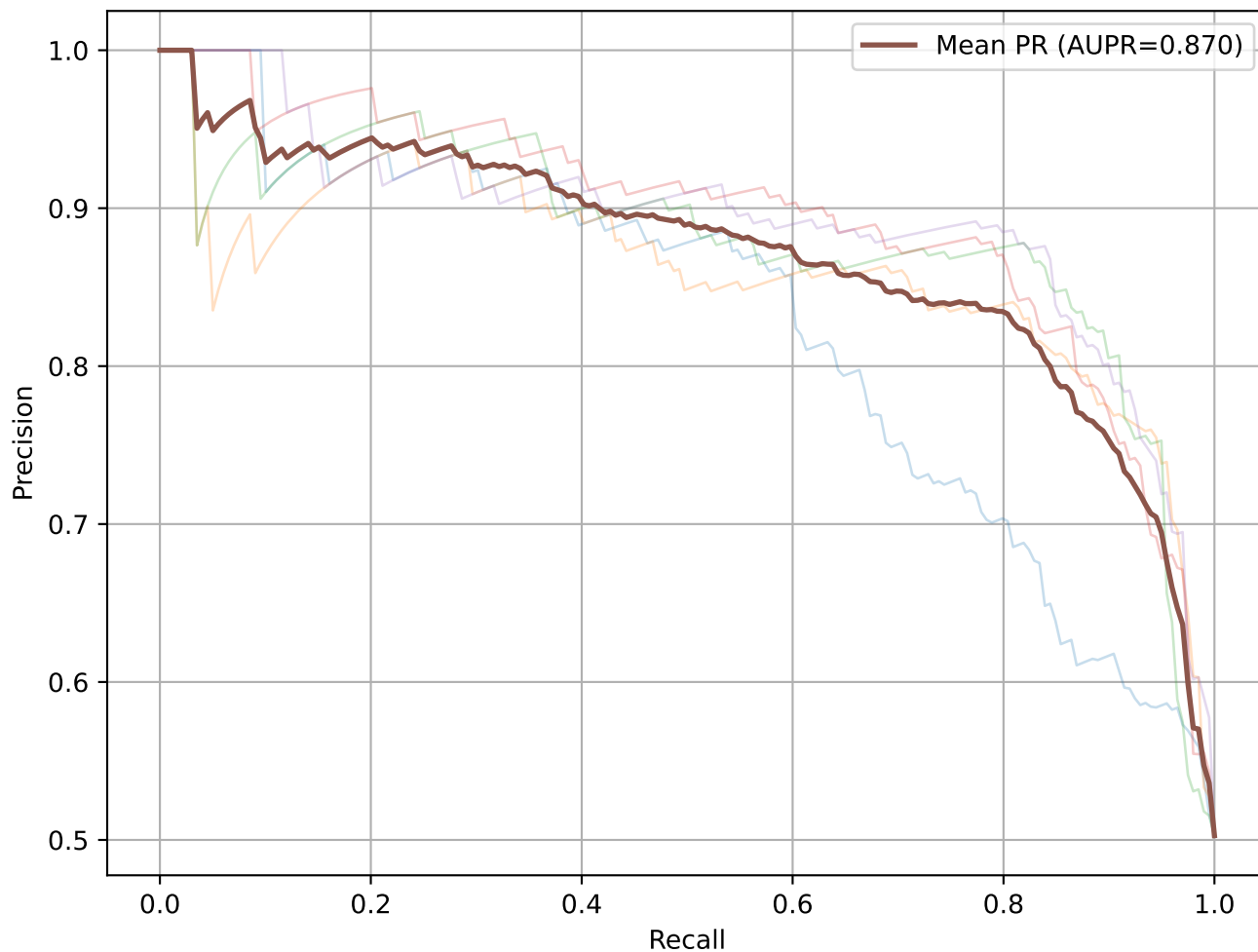
ClassificationReport - RF (last fold)

	precision	recall	f1-score	support
0	0.91	0.86	0.88	199
1	0.87	0.91	0.89	202
accuracy			0.89	401
macro avg	0.89	0.89	0.89	401
weighted avg	0.89	0.89	0.89	401

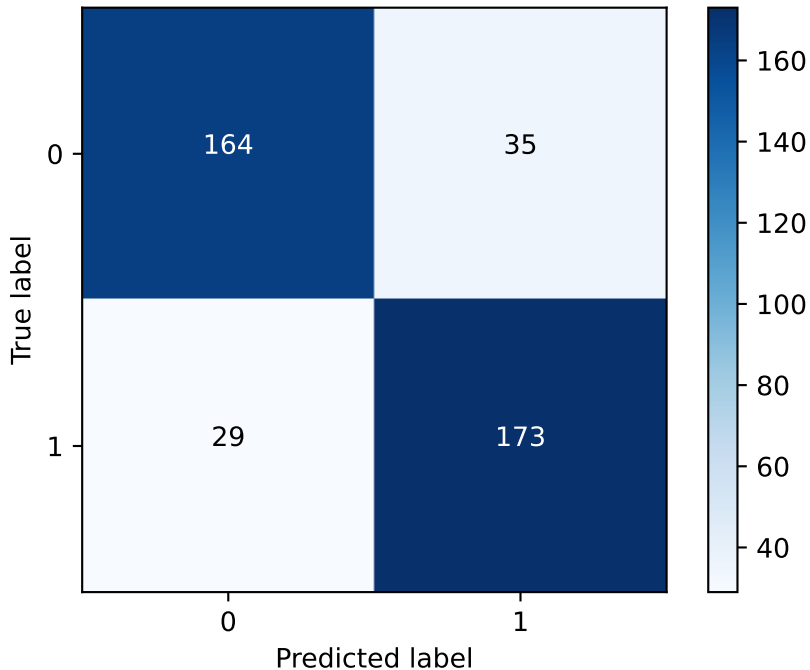
ROC - SVM



Precision-Recall - SVM



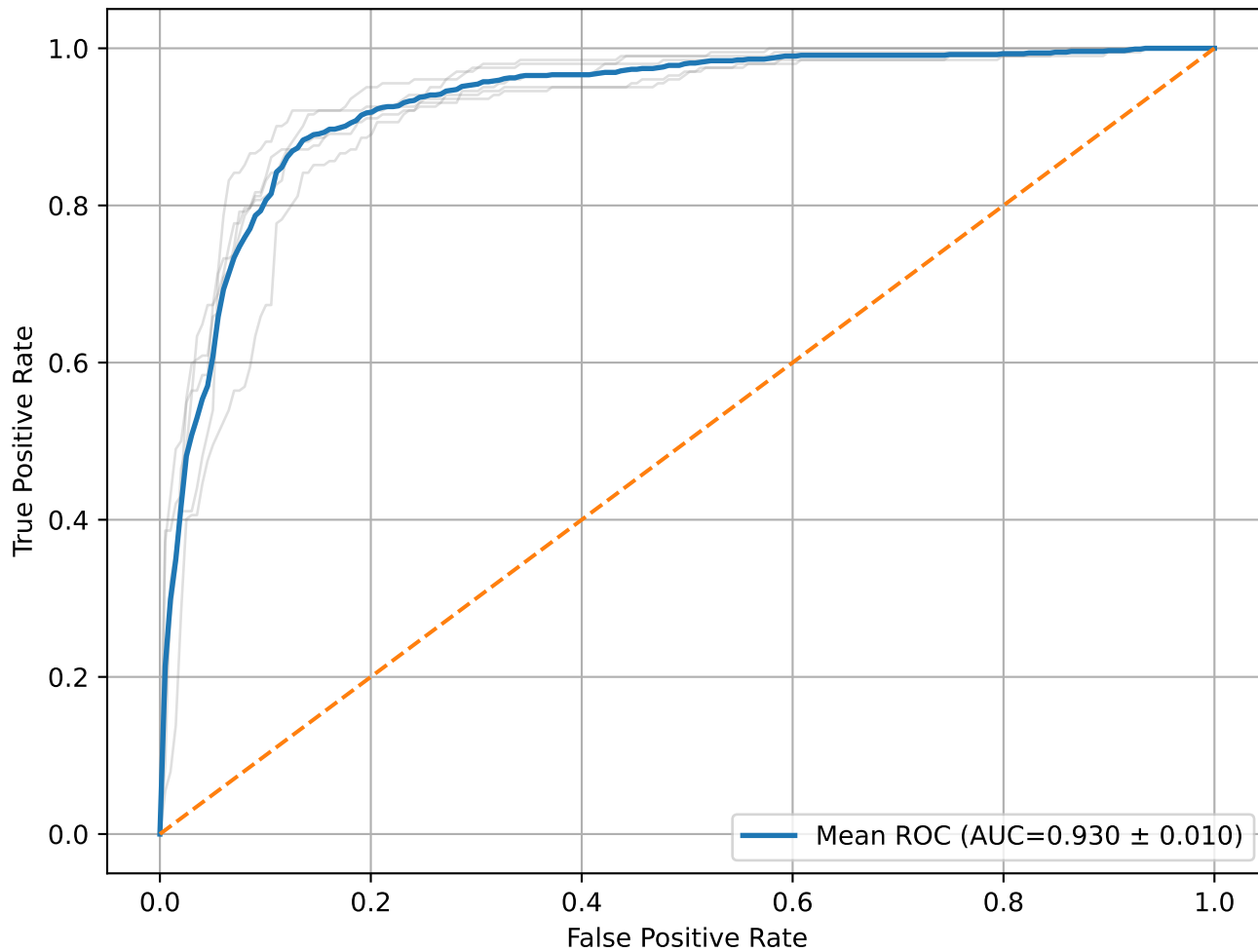
Confusion Matrix - SVM (last fold)



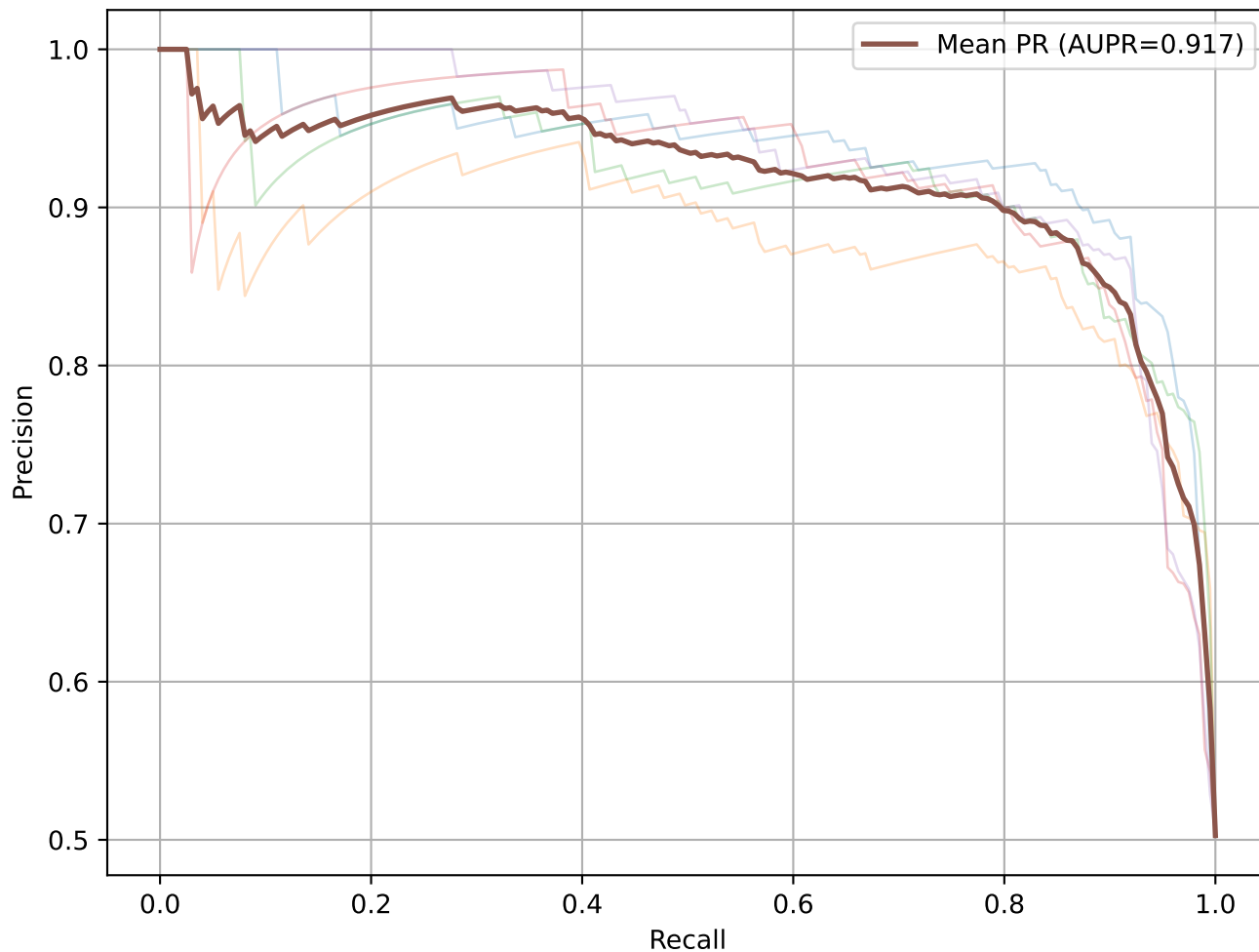
ClassificationReport - SVM (last fold)

	precision	recall	f1-score	support
0	0.85	0.82	0.84	199
1	0.83	0.86	0.84	202
accuracy			0.84	401
macro avg	0.84	0.84	0.84	401
weighted avg	0.84	0.84	0.84	401

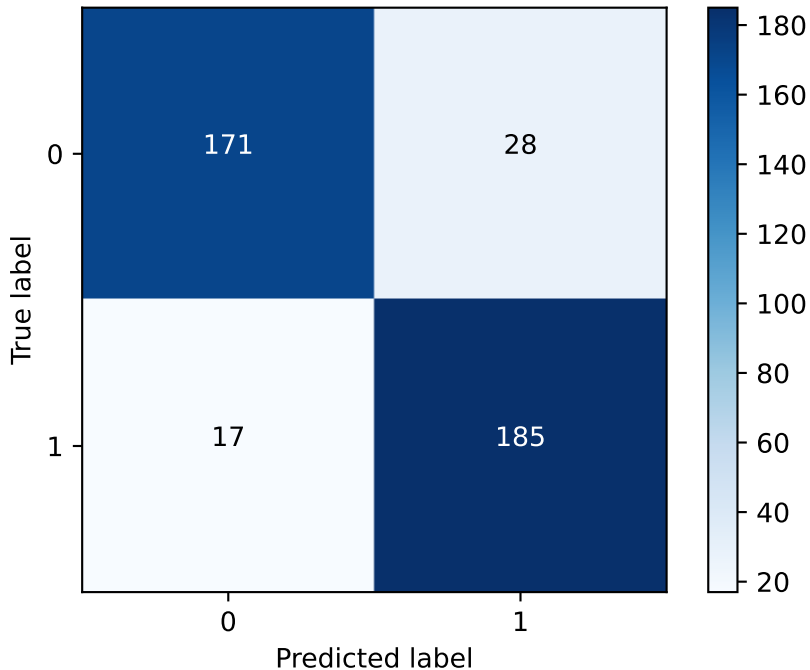
ROC - XGB



Precision-Recall - XGB



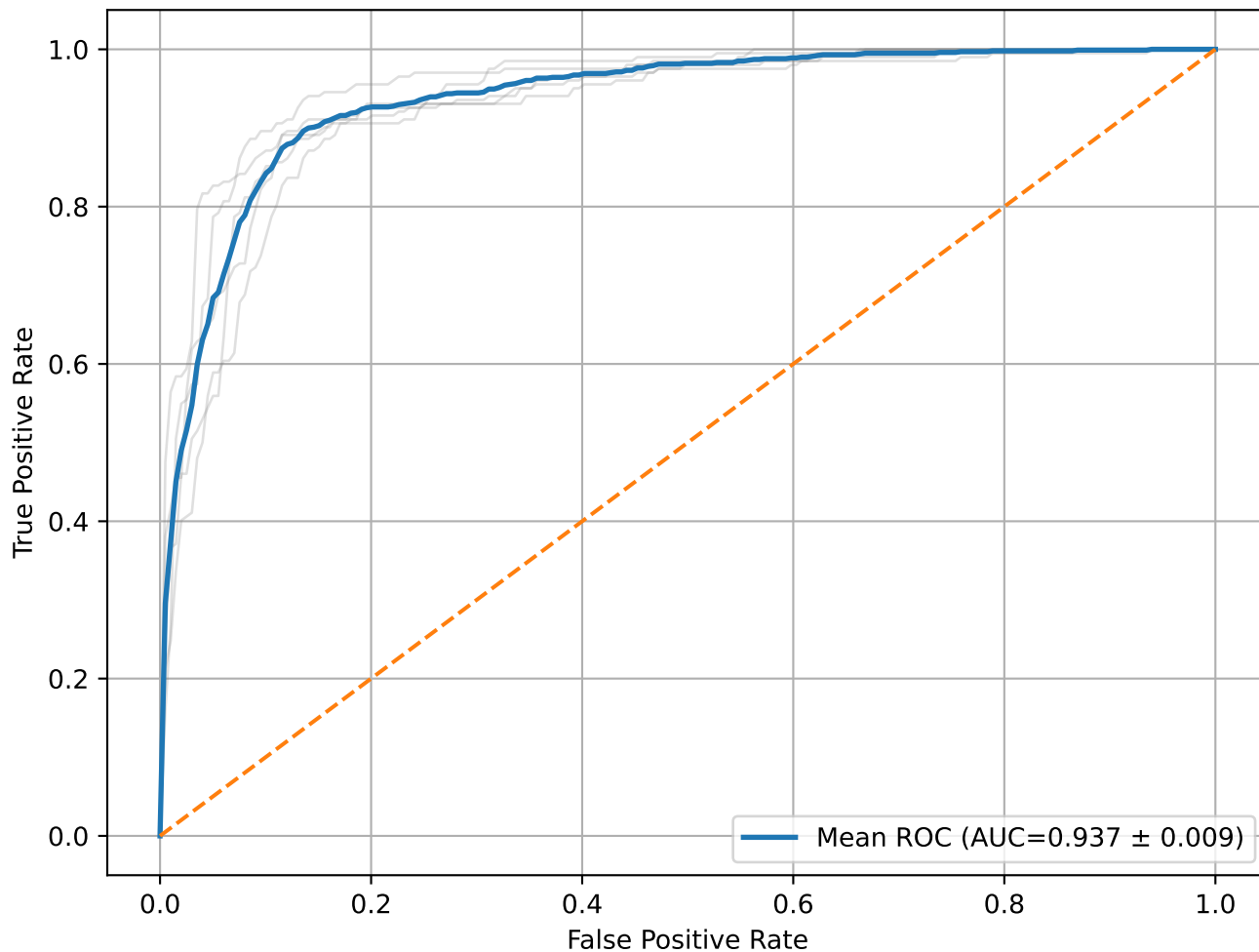
Confusion Matrix - XGB (last fold)



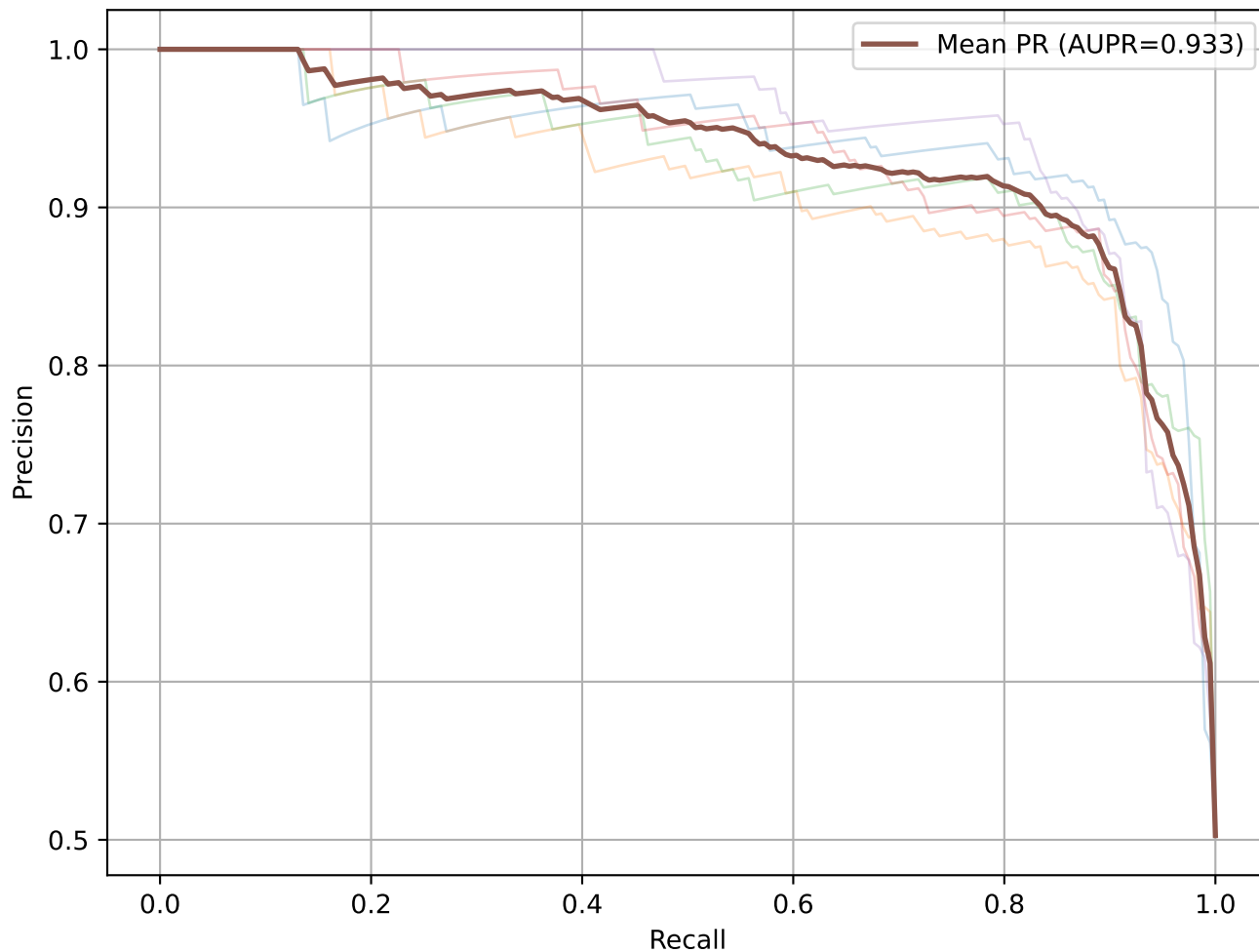
ClassificationReport - XGB (last fold)

	precision	recall	f1-score	support
0	0.91	0.86	0.88	199
1	0.87	0.92	0.89	202
accuracy			0.89	401
macro avg	0.89	0.89	0.89	401
weighted avg	0.89	0.89	0.89	401

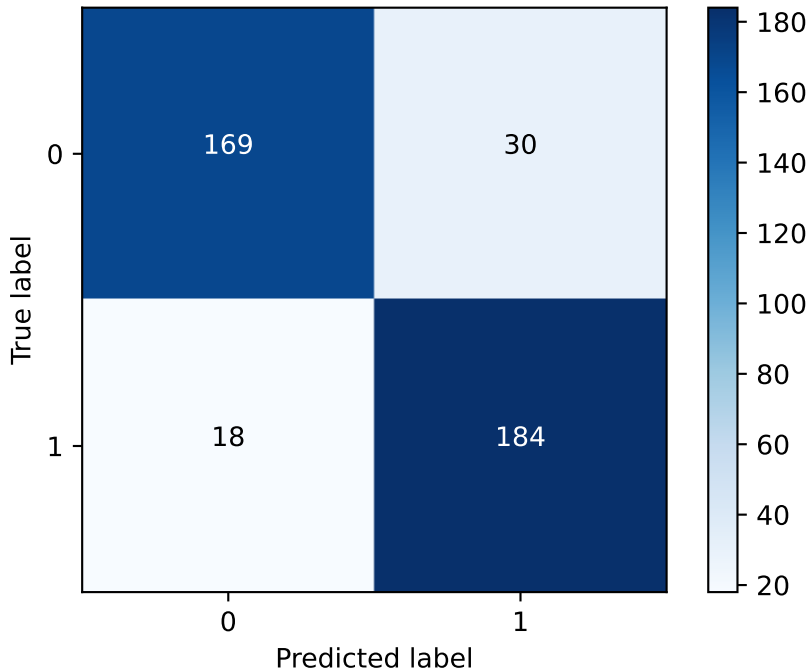
ROC - LGB



Precision-Recall - LGB



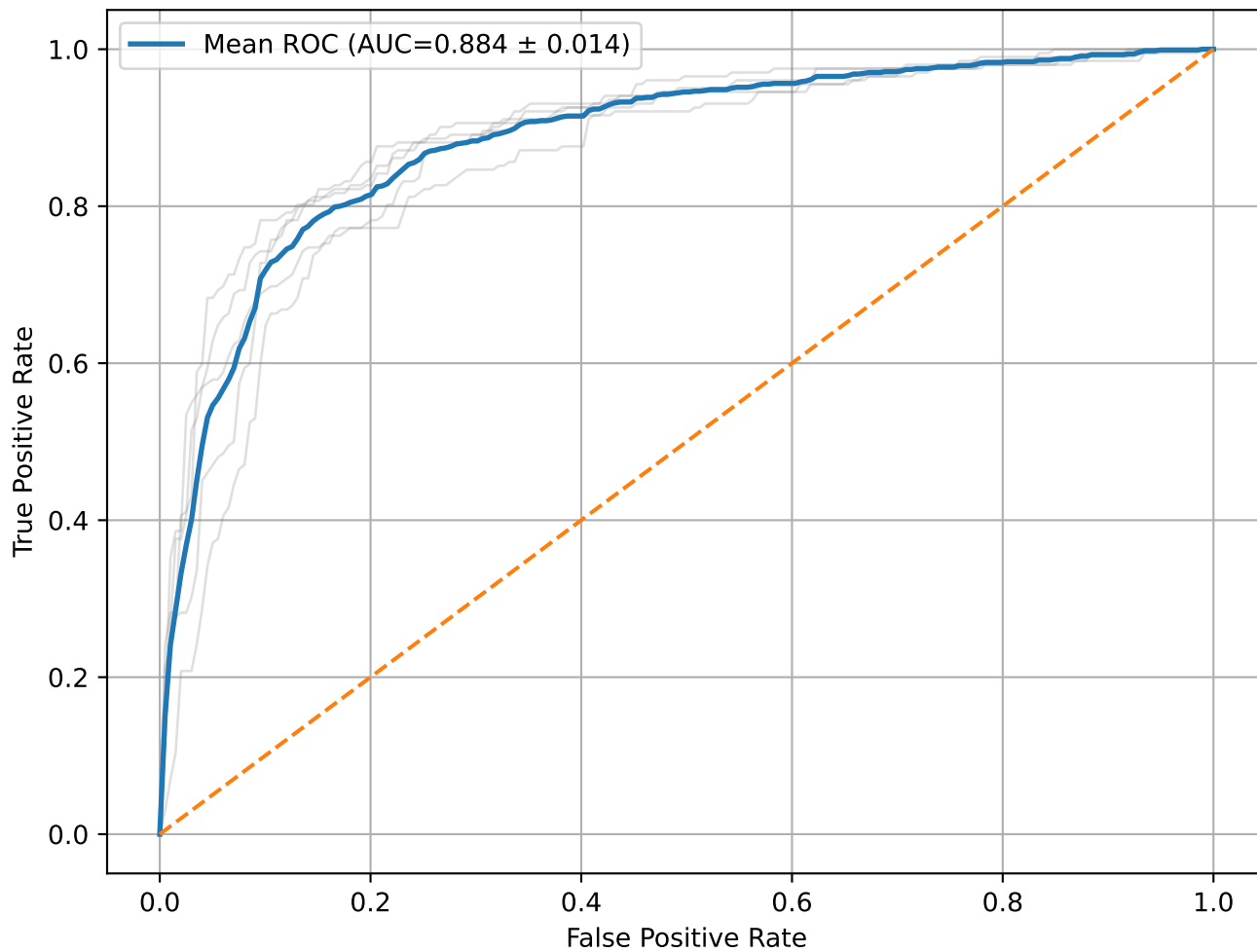
Confusion Matrix - LGB (last fold)



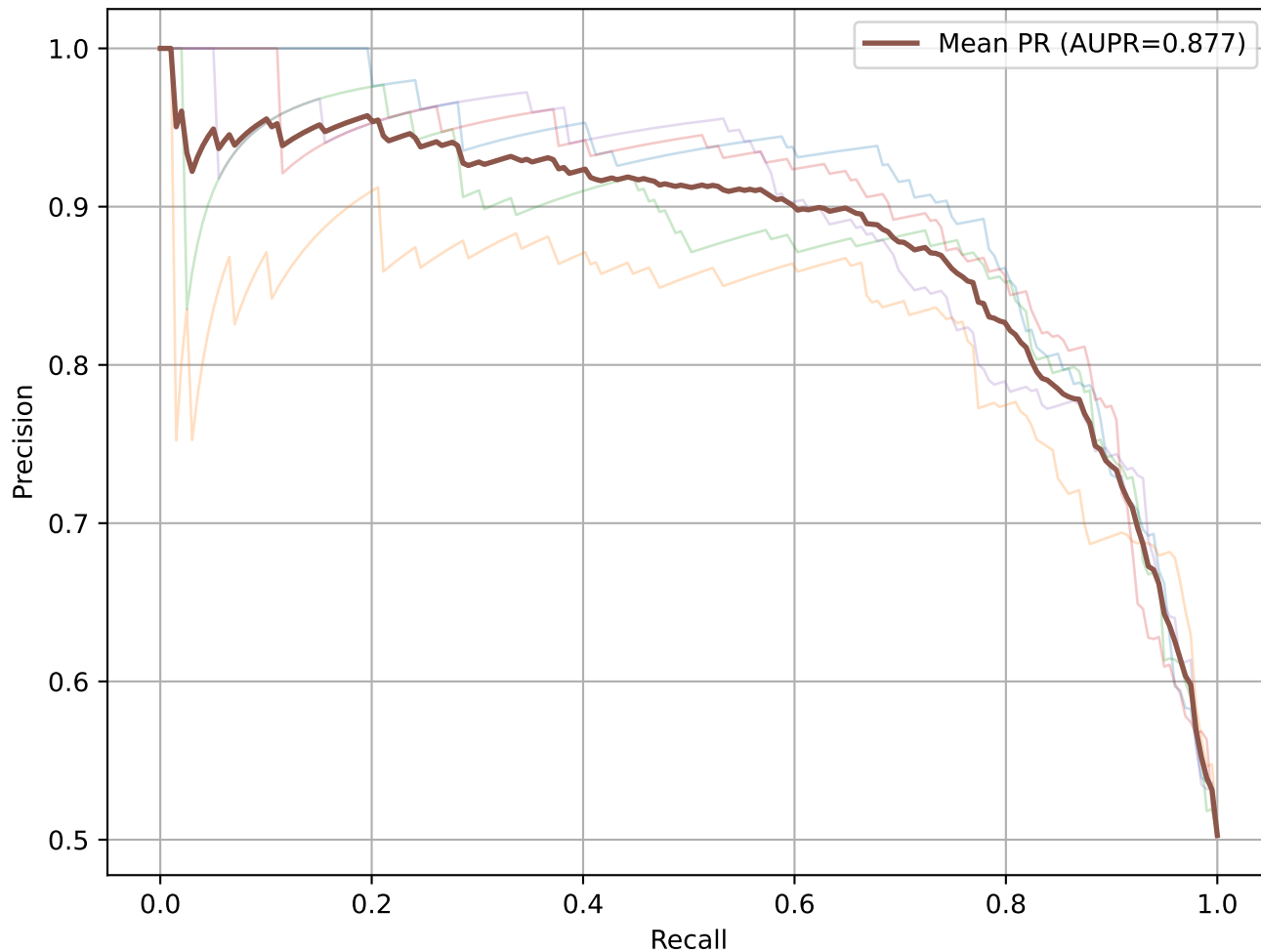
ClassificationReport - LGB (last fold)

	precision	recall	f1-score	support
0	0.90	0.85	0.88	199
1	0.86	0.91	0.88	202
accuracy			0.88	401
macro avg	0.88	0.88	0.88	401
weighted avg	0.88	0.88	0.88	401

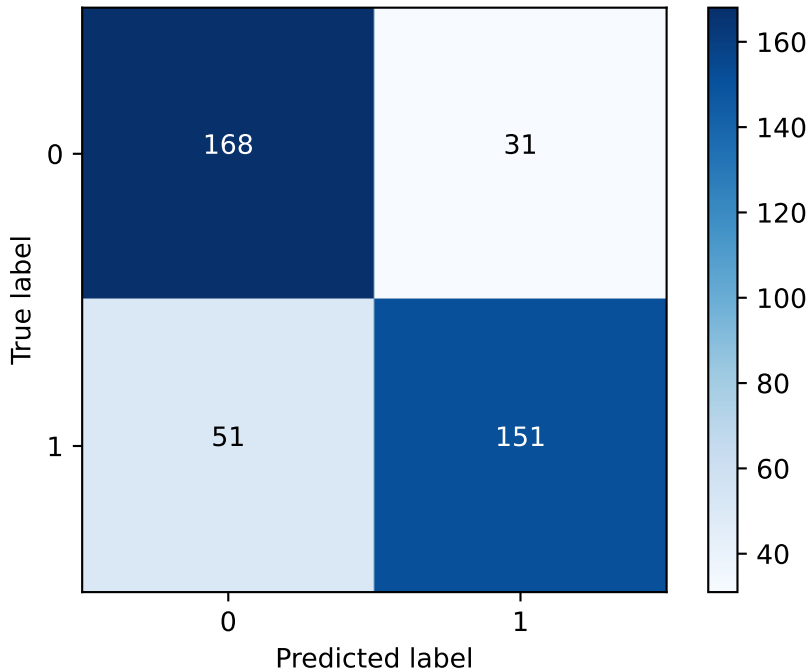
ROC - DNN



Precision-Recall - DNN



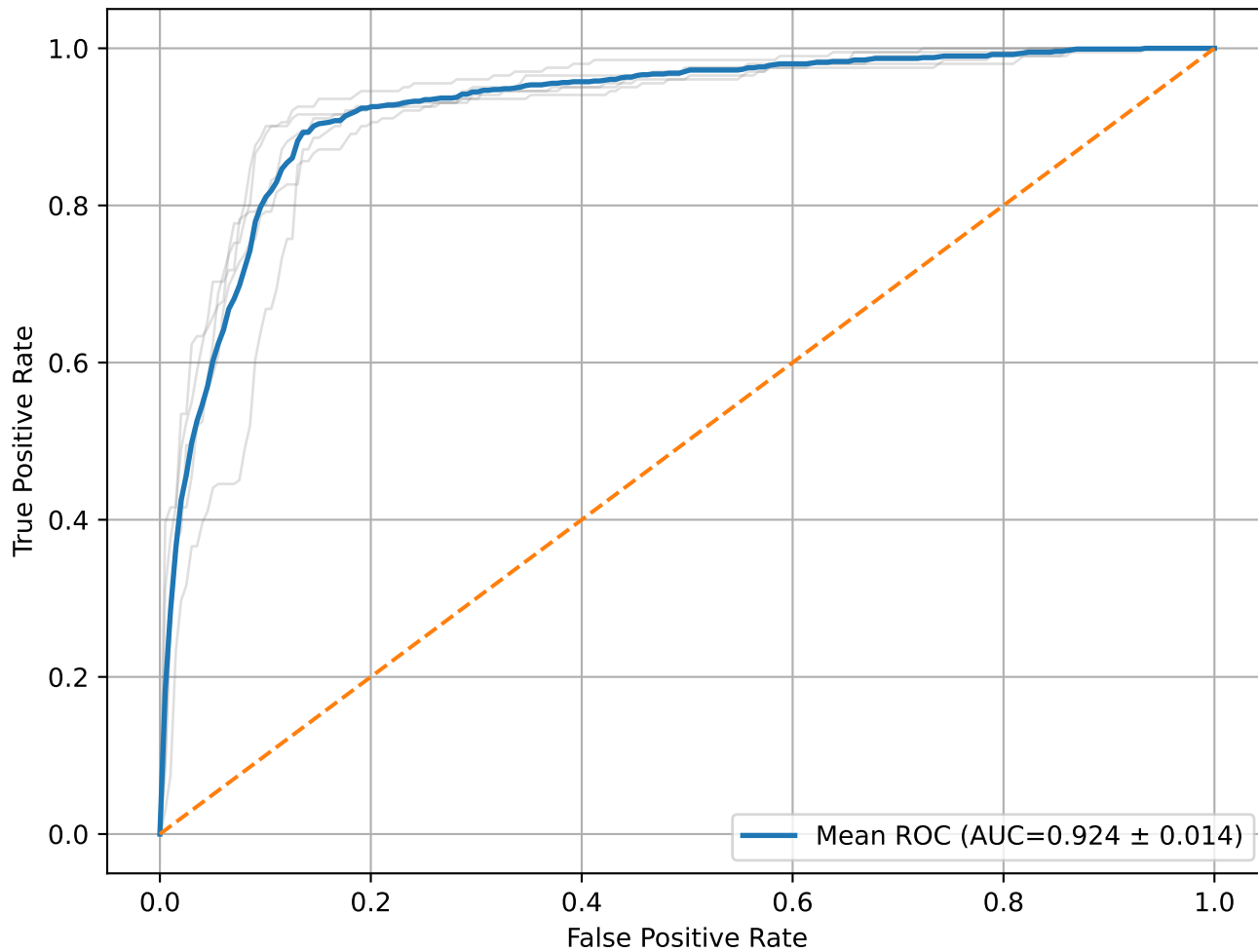
Confusion Matrix - DNN (last fold)



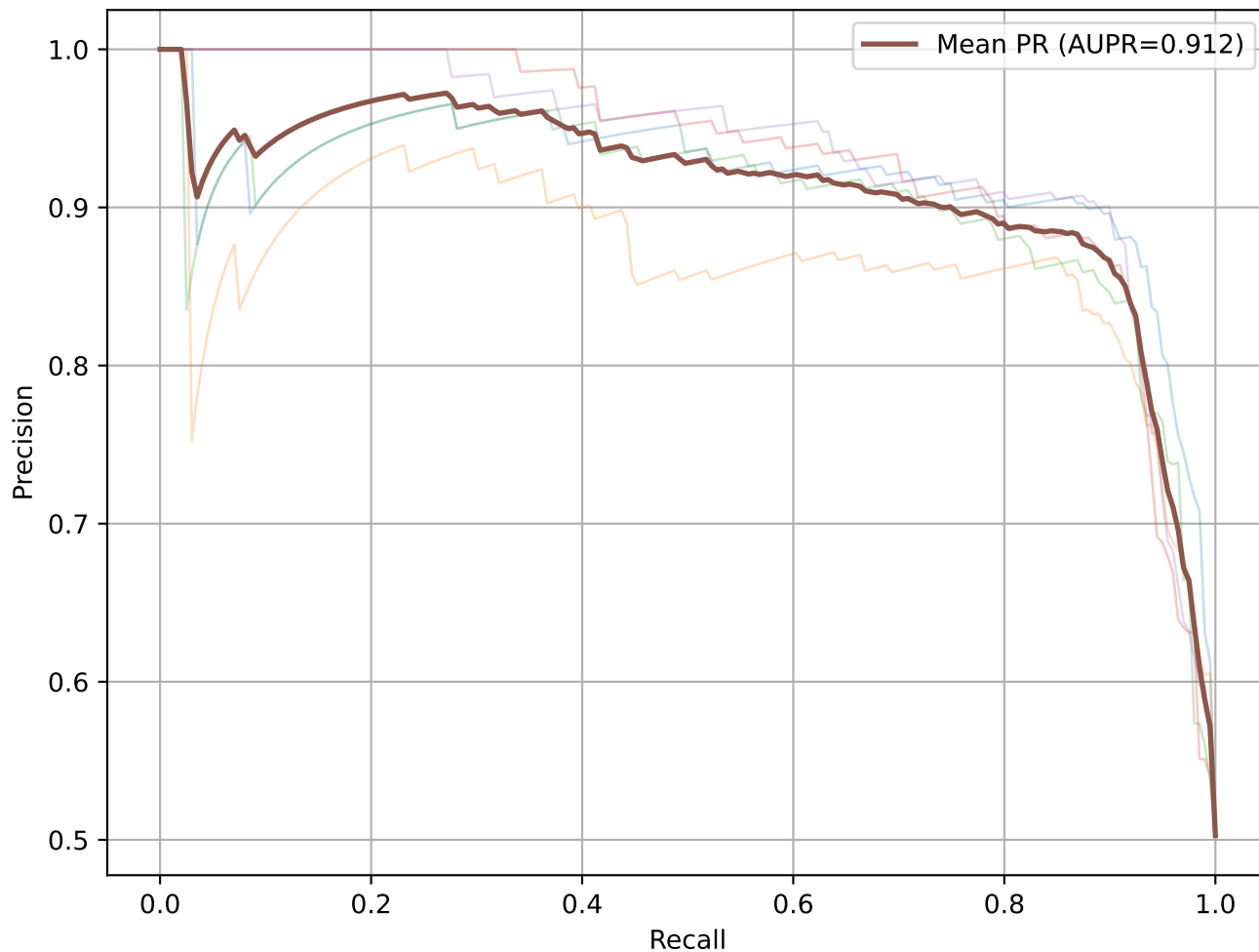
ClassificationReport - DNN (last fold)

	precision	recall	f1-score	support
0	0.77	0.84	0.80	199
1	0.83	0.75	0.79	202
accuracy			0.80	401
macro avg	0.80	0.80	0.80	401
weighted avg	0.80	0.80	0.80	401

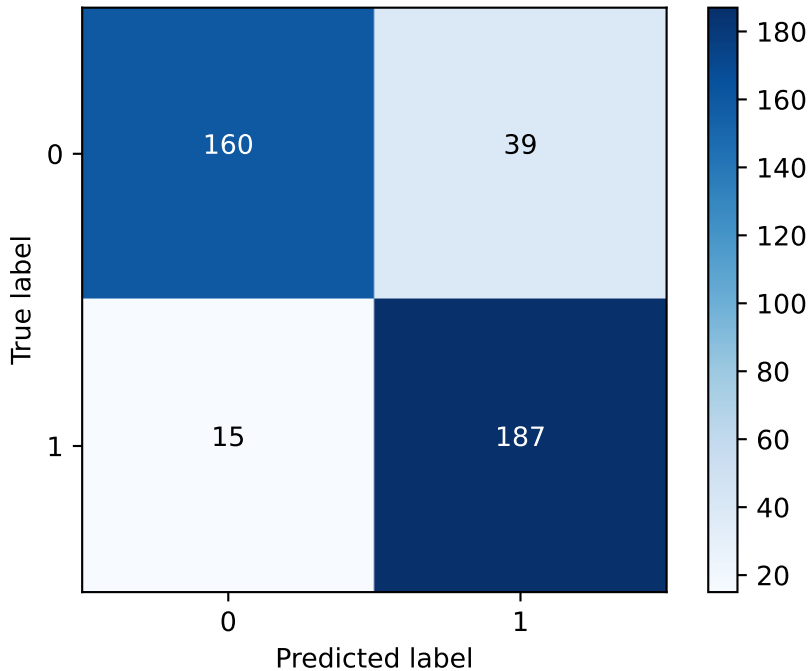
ROC - STACKING



Precision-Recall - STACKING



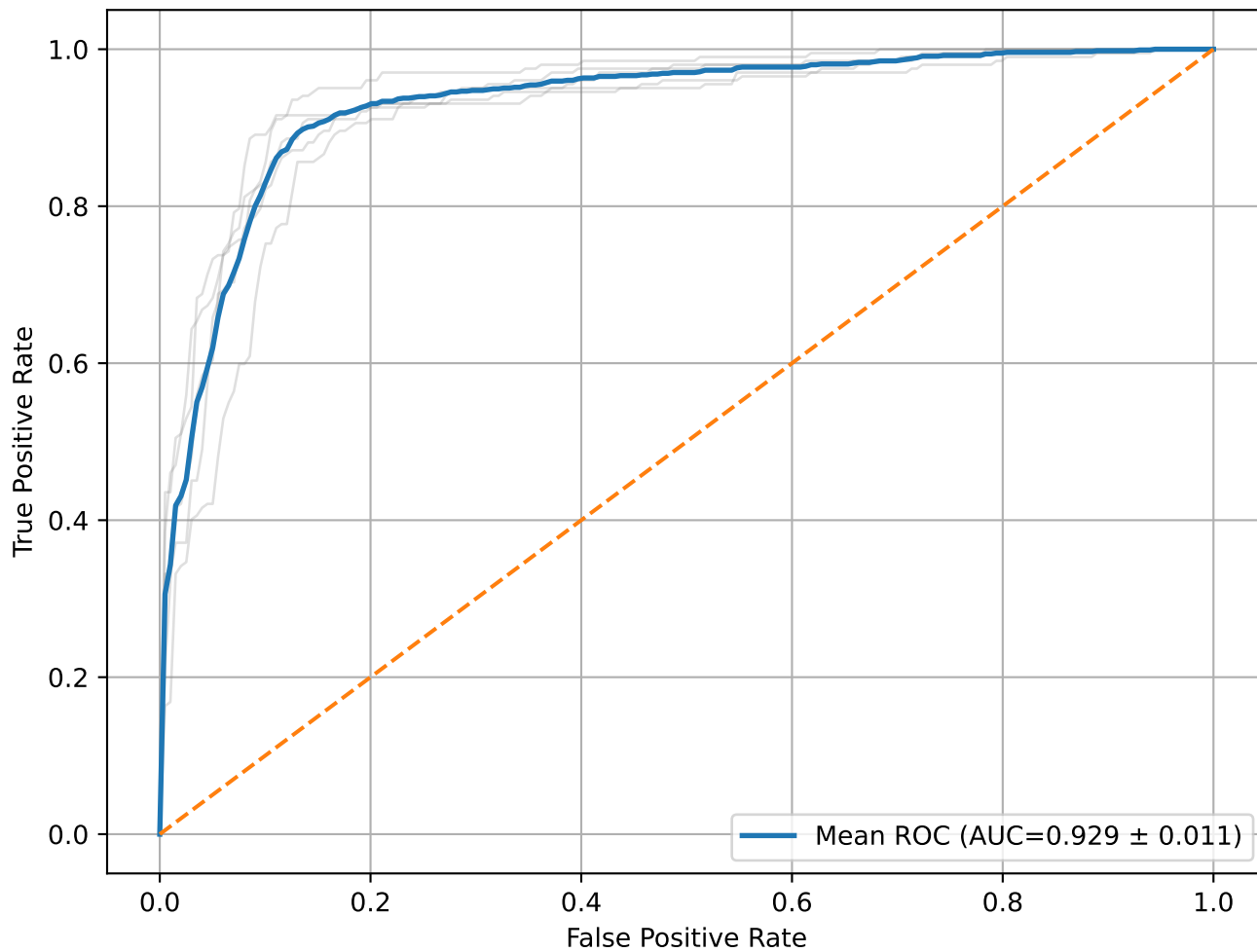
Confusion Matrix - STACKING (last fold)

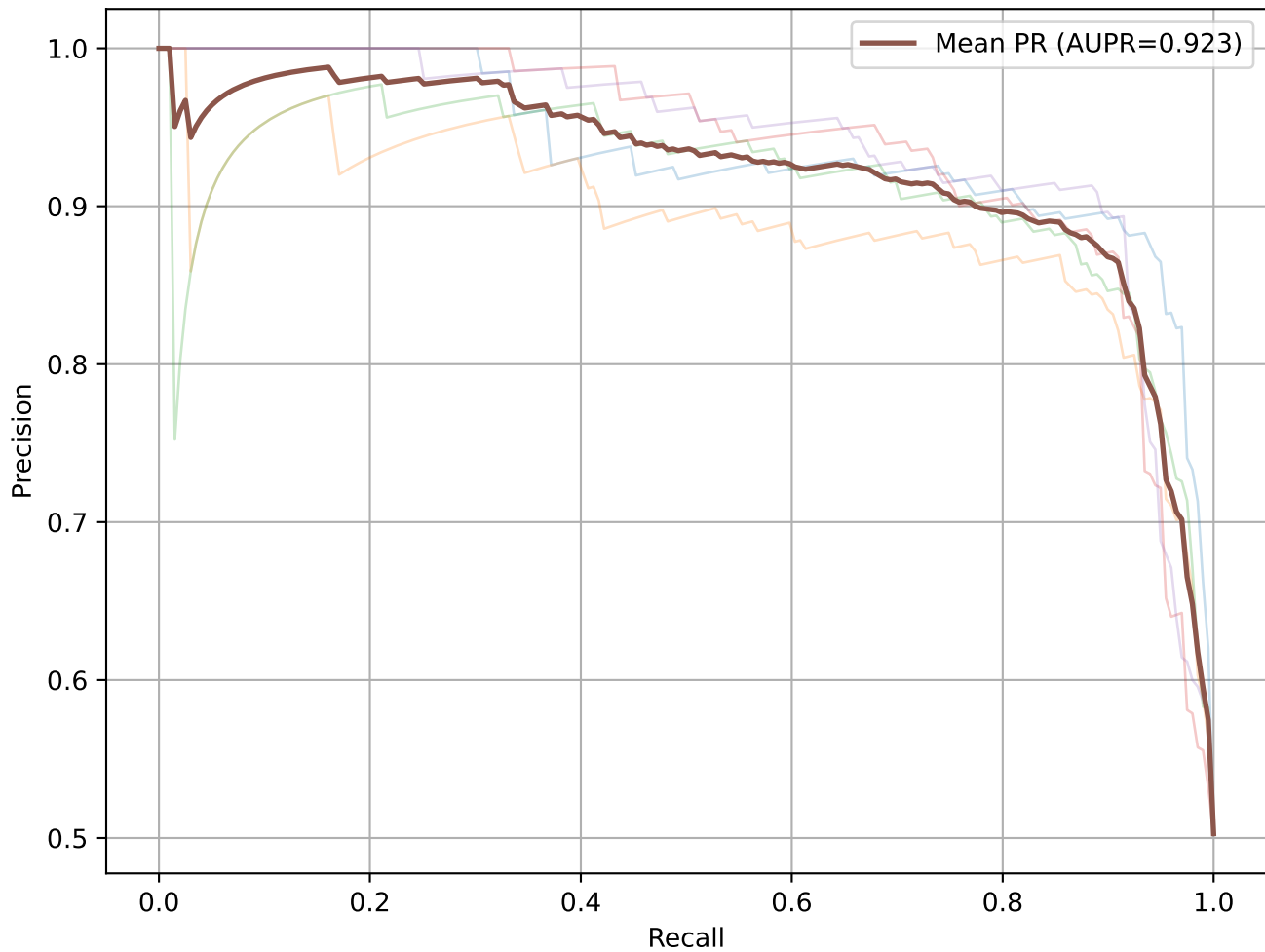


ClassificationReport - STACKING (last fold)

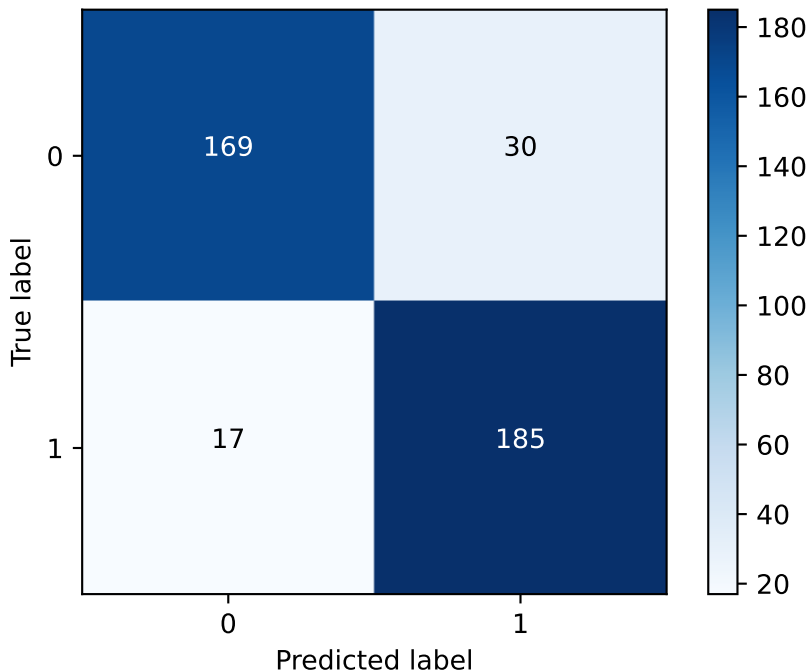
	precision	recall	f1-score	support
0	0.91	0.80	0.86	199
1	0.83	0.93	0.87	202
accuracy			0.87	401
macro avg	0.87	0.86	0.86	401
weighted avg	0.87	0.87	0.86	401

ROC - VOTING





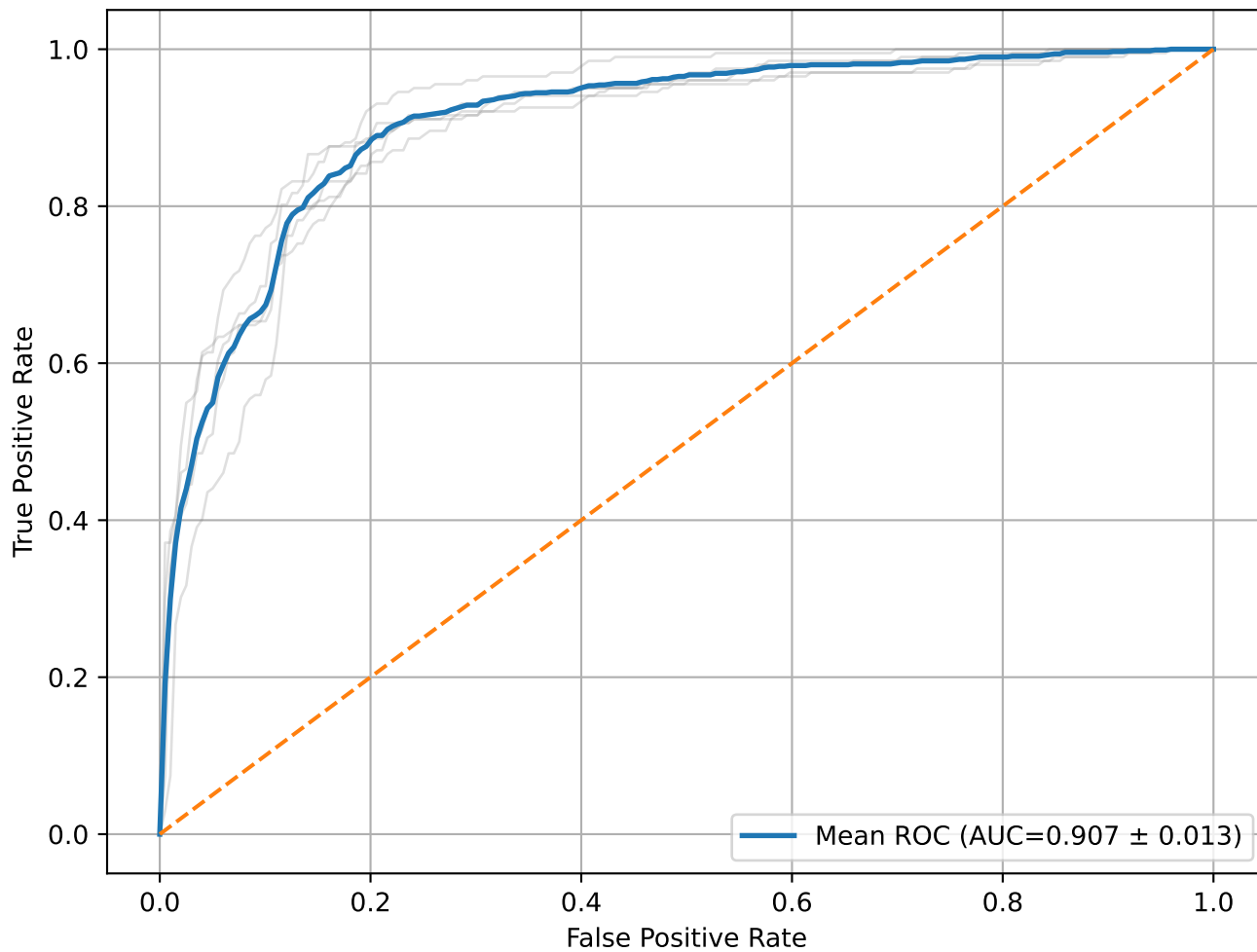
Confusion Matrix - VOTING (last fold)



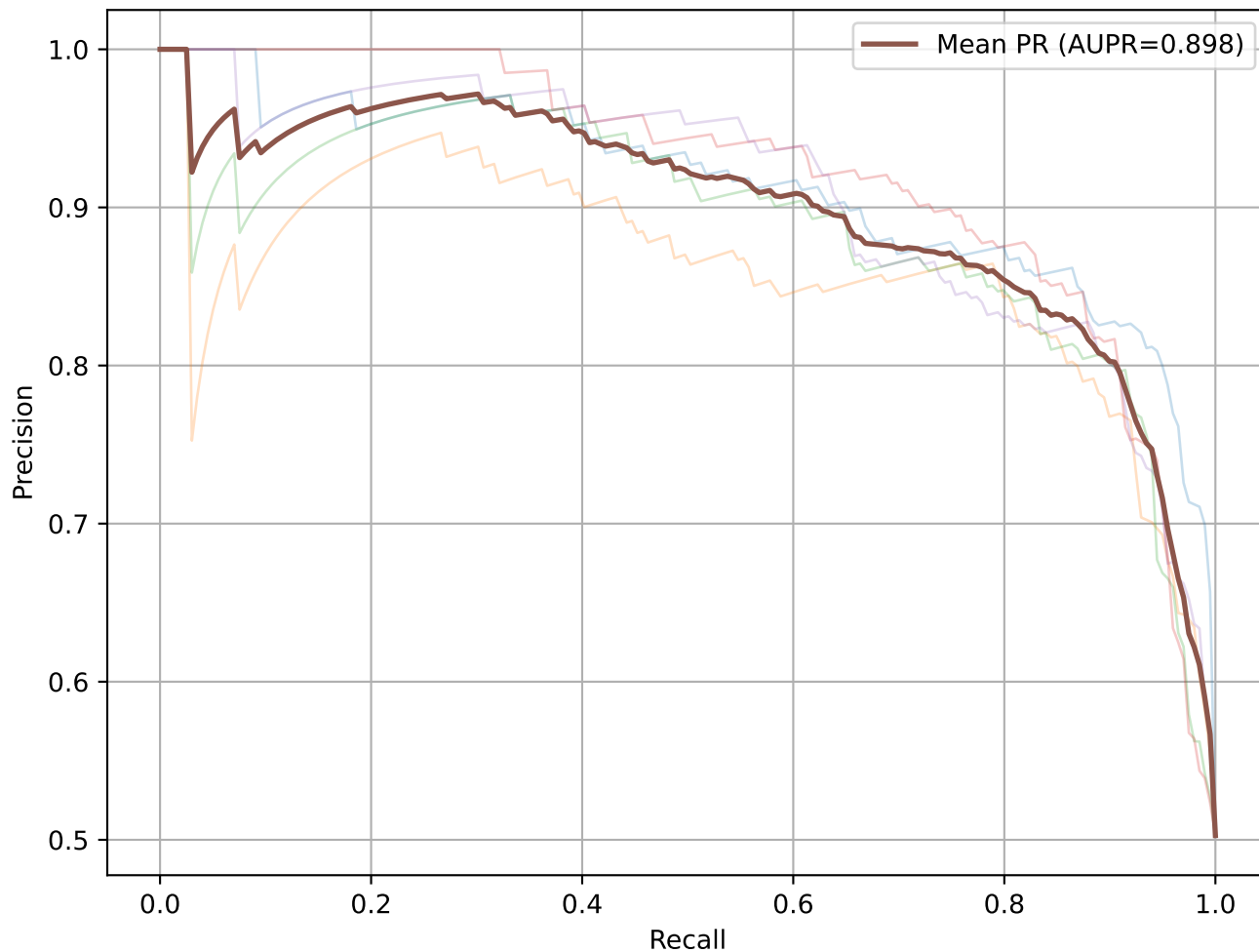
ClassificationReport - VOTING (last fold)

	precision	recall	f1-score	support
0	0.91	0.85	0.88	199
1	0.86	0.92	0.89	202
accuracy			0.88	401
macro avg	0.88	0.88	0.88	401
weighted avg	0.88	0.88	0.88	401

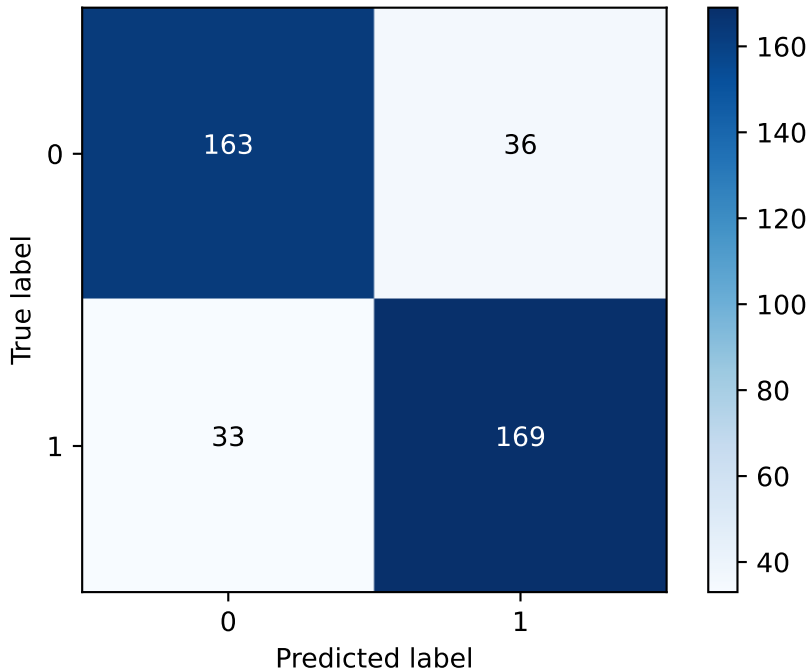
ROC - BAGGING



Precision-Recall - BAGGING



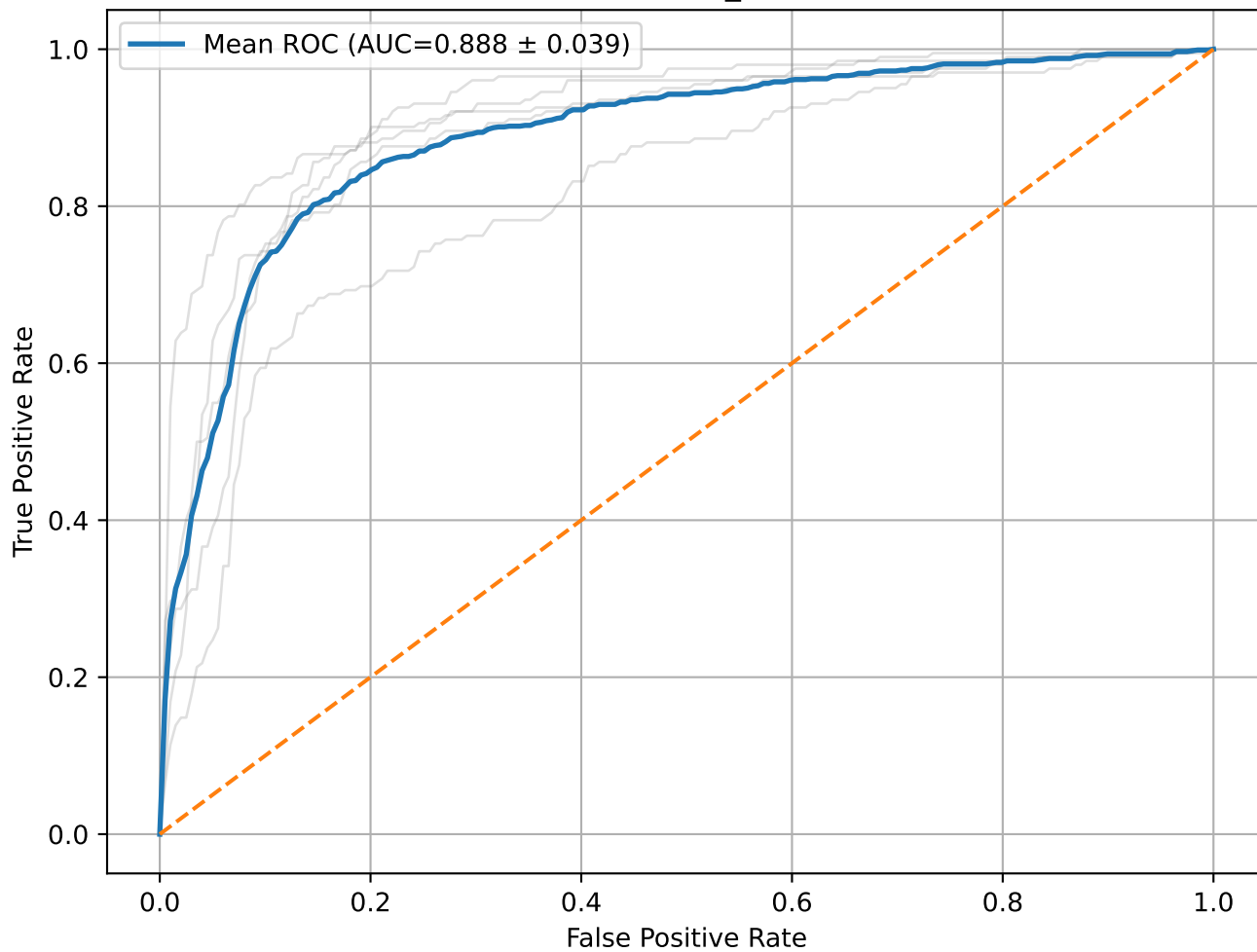
Confusion Matrix - BAGGING (last fold)



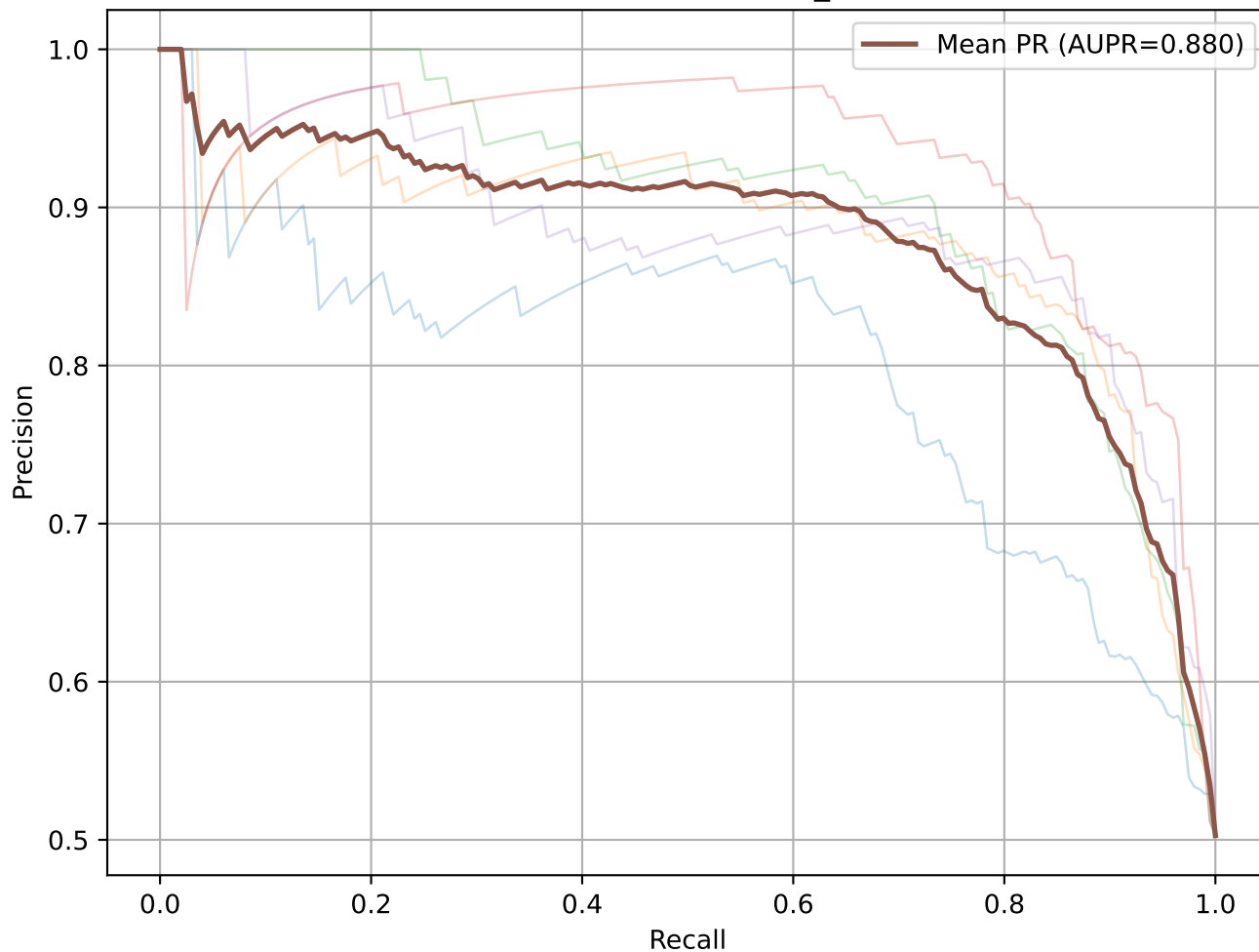
ClassificationReport - BAGGING (last fold)

	precision	recall	f1-score	support
0	0.83	0.82	0.83	199
1	0.82	0.84	0.83	202
accuracy			0.83	401
macro avg	0.83	0.83	0.83	401
weighted avg	0.83	0.83	0.83	401

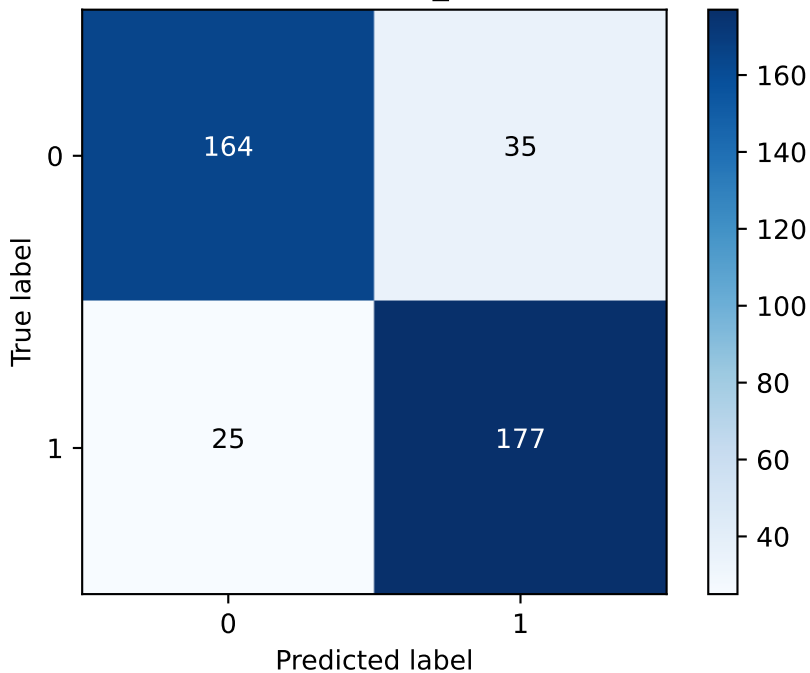
ROC - PCA_SVM



Precision-Recall - PCA_SVM



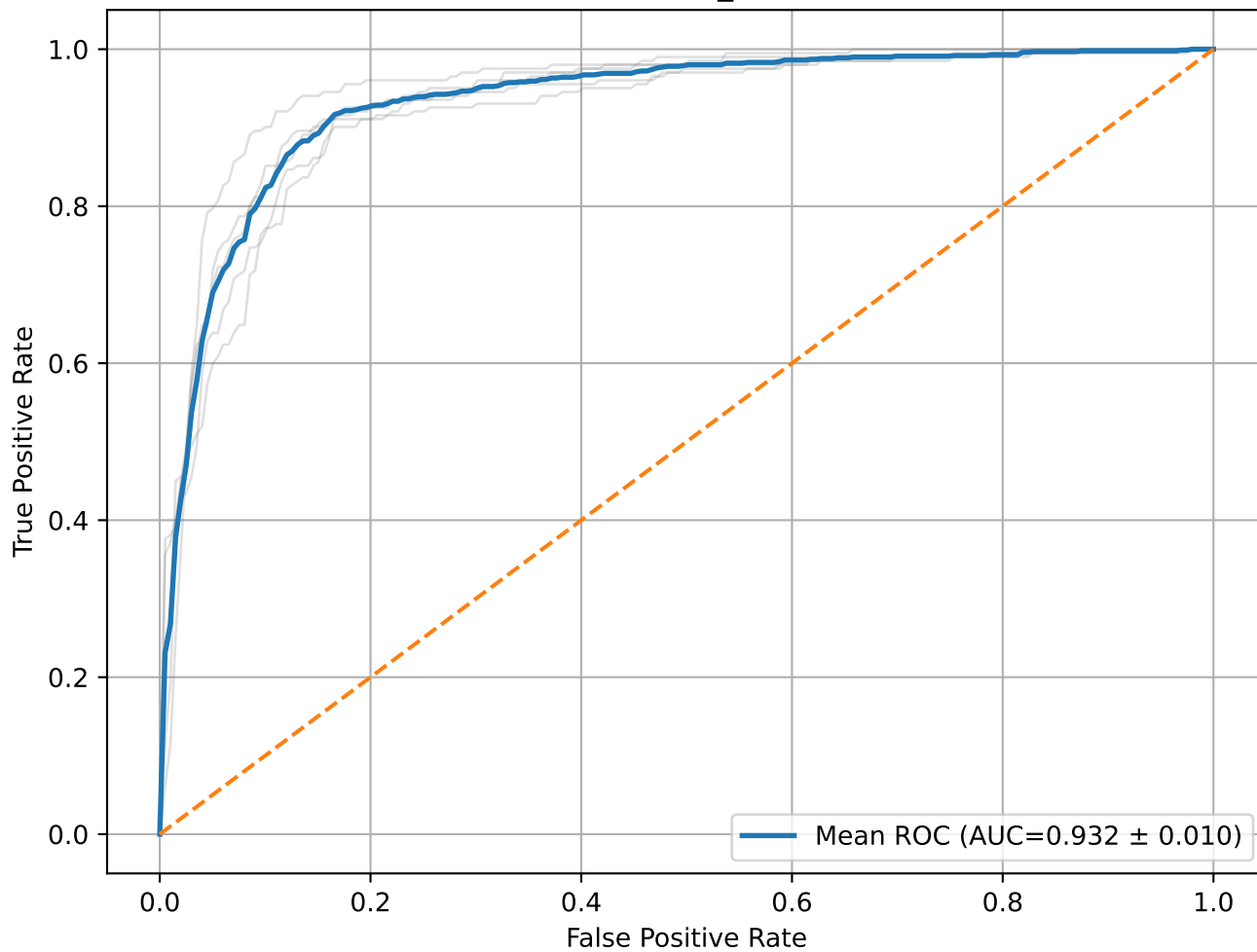
Confusion Matrix - PCA_SVM (last fold)



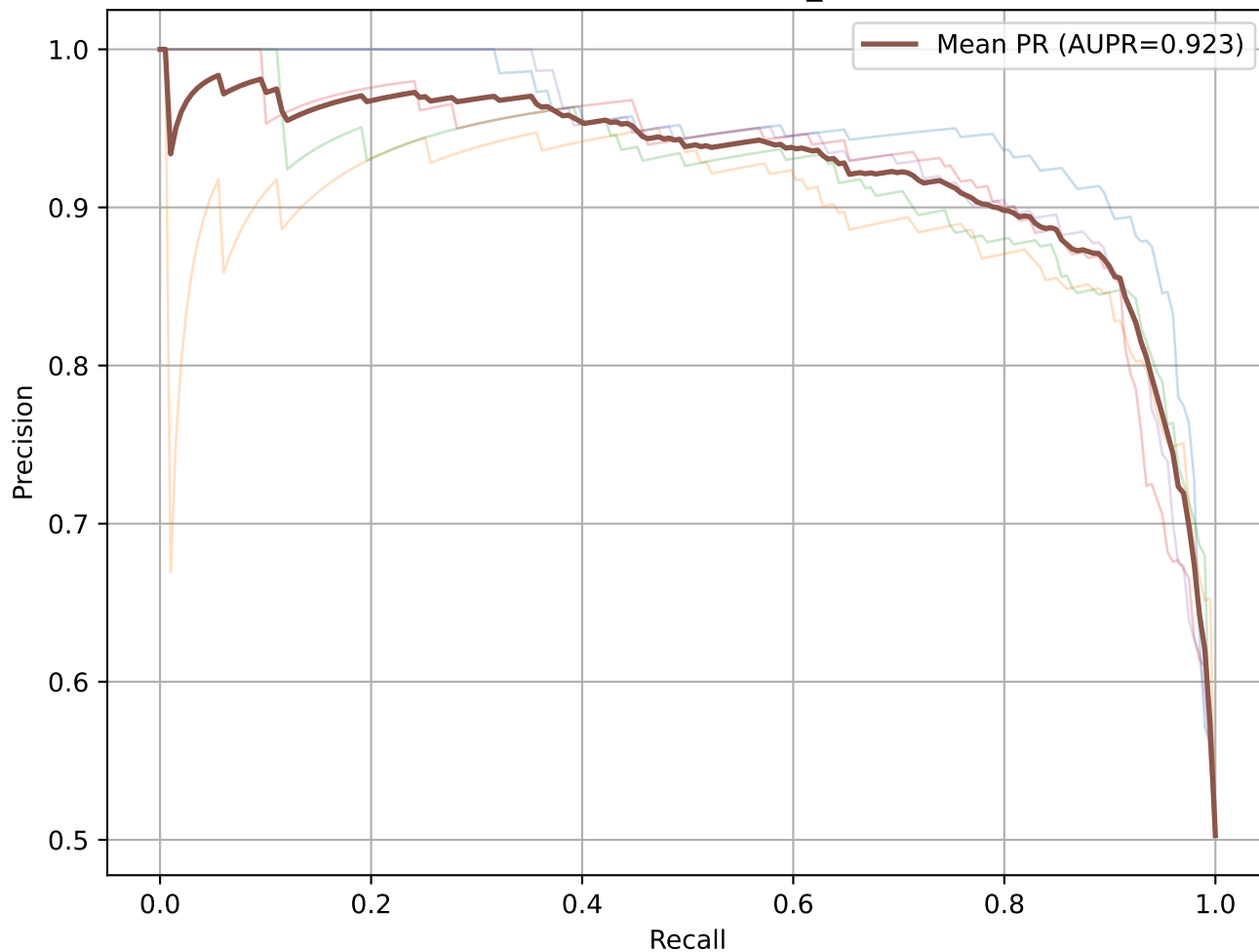
ClassificationReport - PCA_SVM (last fold)

	precision	recall	f1-score	support
0	0.87	0.82	0.85	199
1	0.83	0.88	0.86	202
accuracy			0.85	401
macro avg	0.85	0.85	0.85	401
weighted avg	0.85	0.85	0.85	401

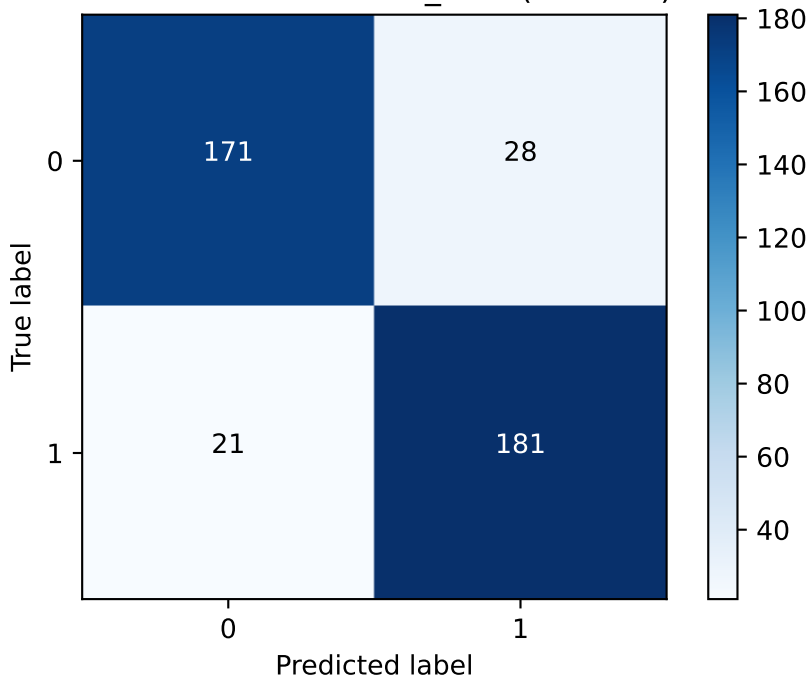
ROC - GA_XGB



Precision-Recall - GA_XGB



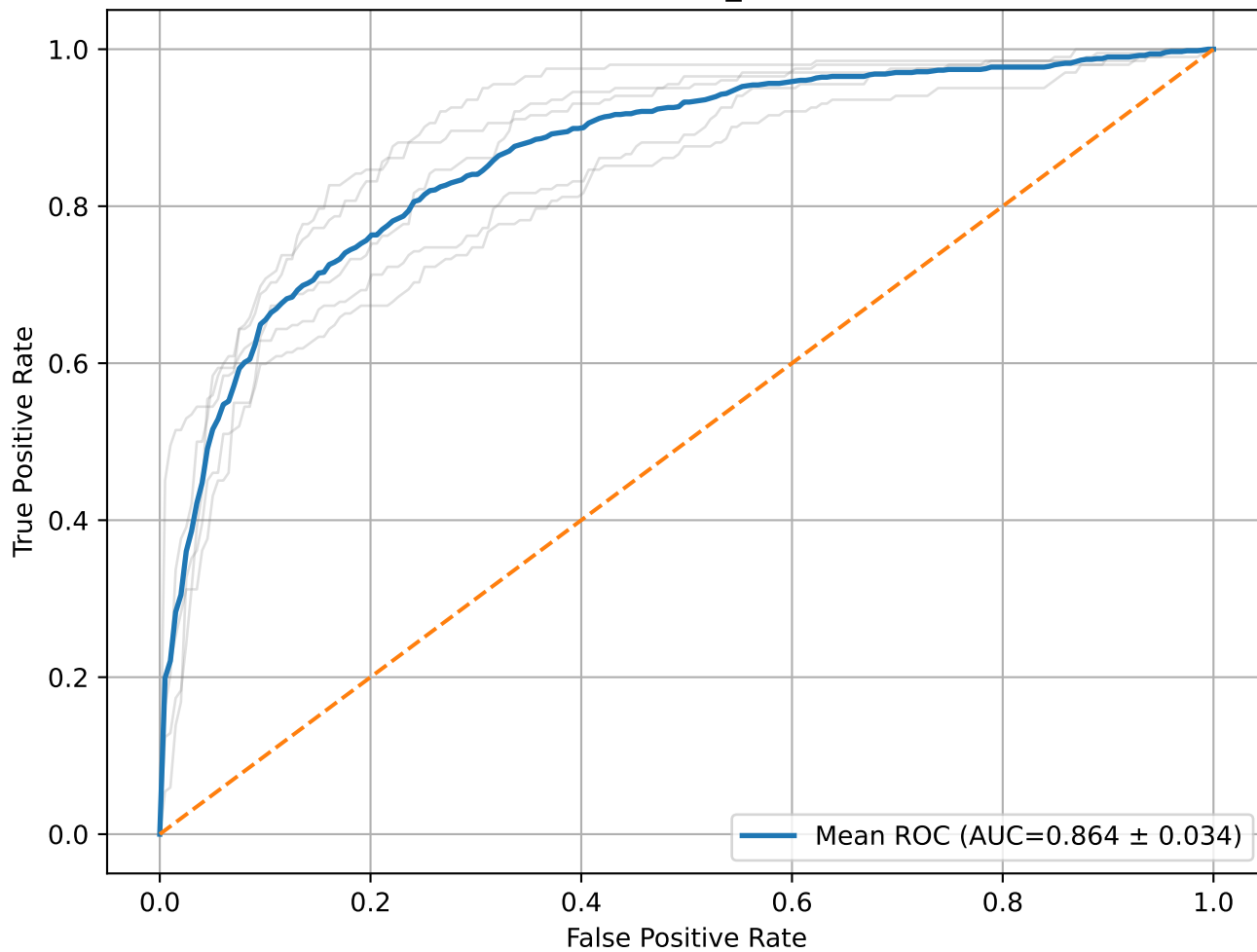
Confusion Matrix - GA_XGB (last fold)



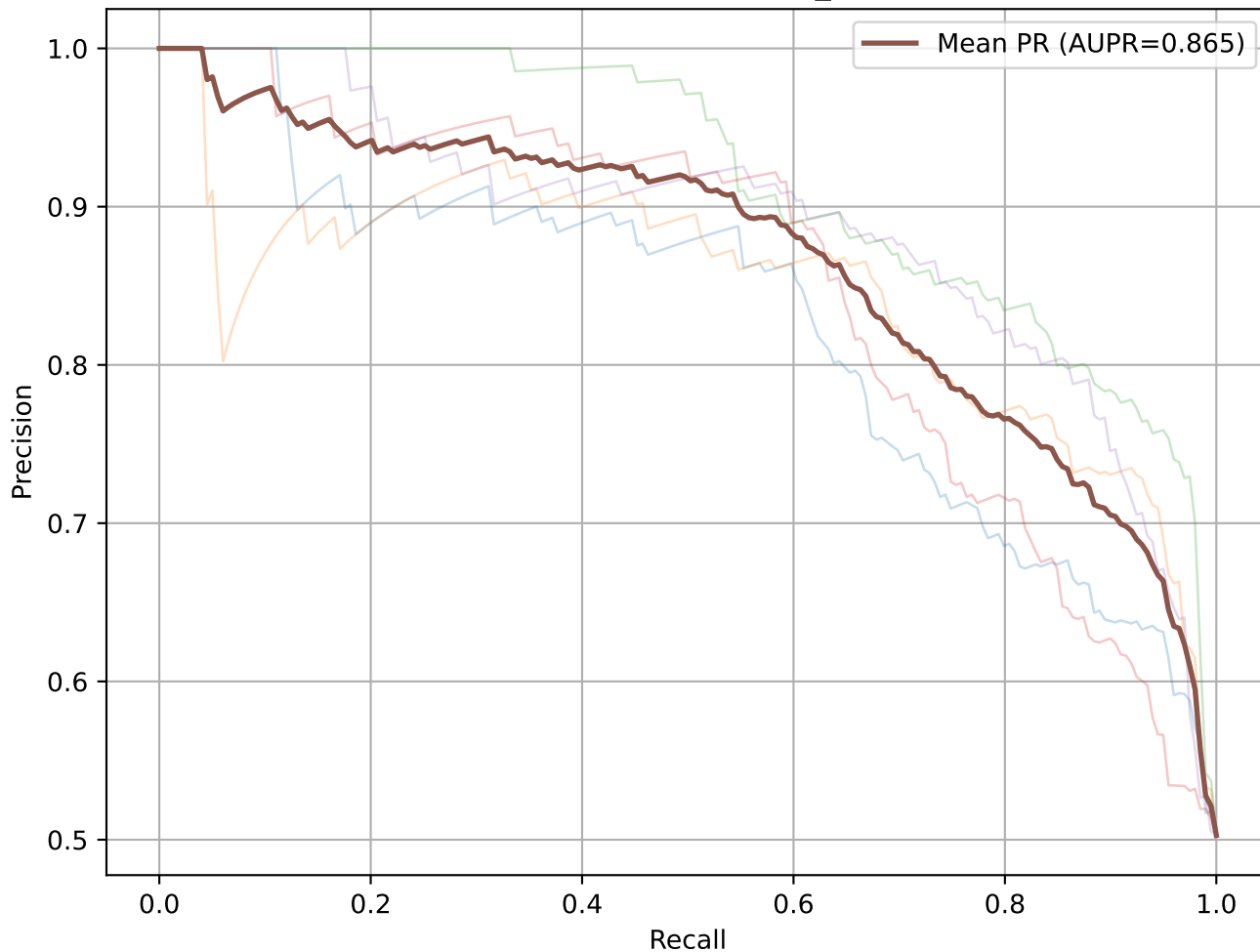
ClassificationReport - GA_XGB (last fold)

	precision	recall	f1-score	support
0	0.89	0.86	0.87	199
1	0.87	0.90	0.88	202
accuracy			0.88	401
macro avg	0.88	0.88	0.88	401
weighted avg	0.88	0.88	0.88	401

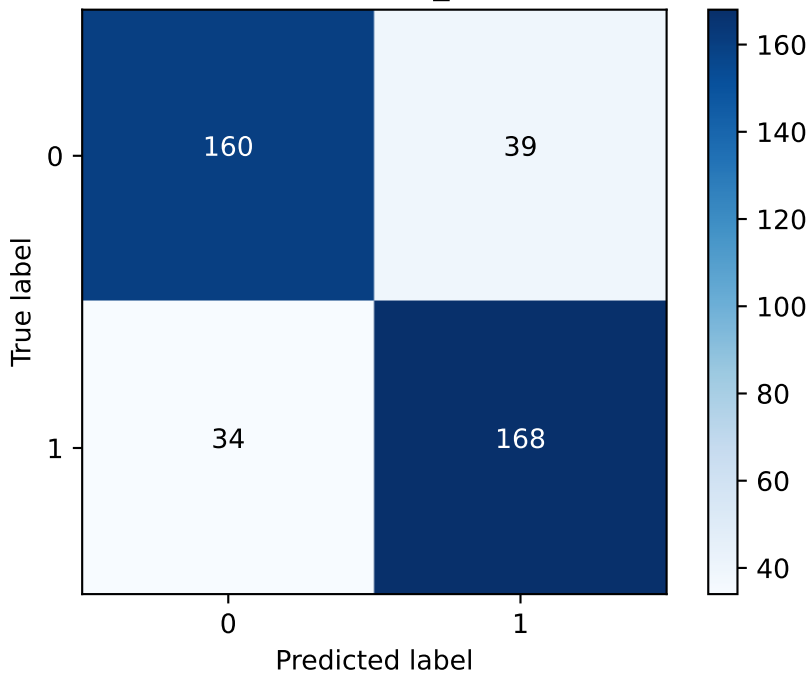
ROC - PSO_SVM



Precision-Recall - PSO_SVM



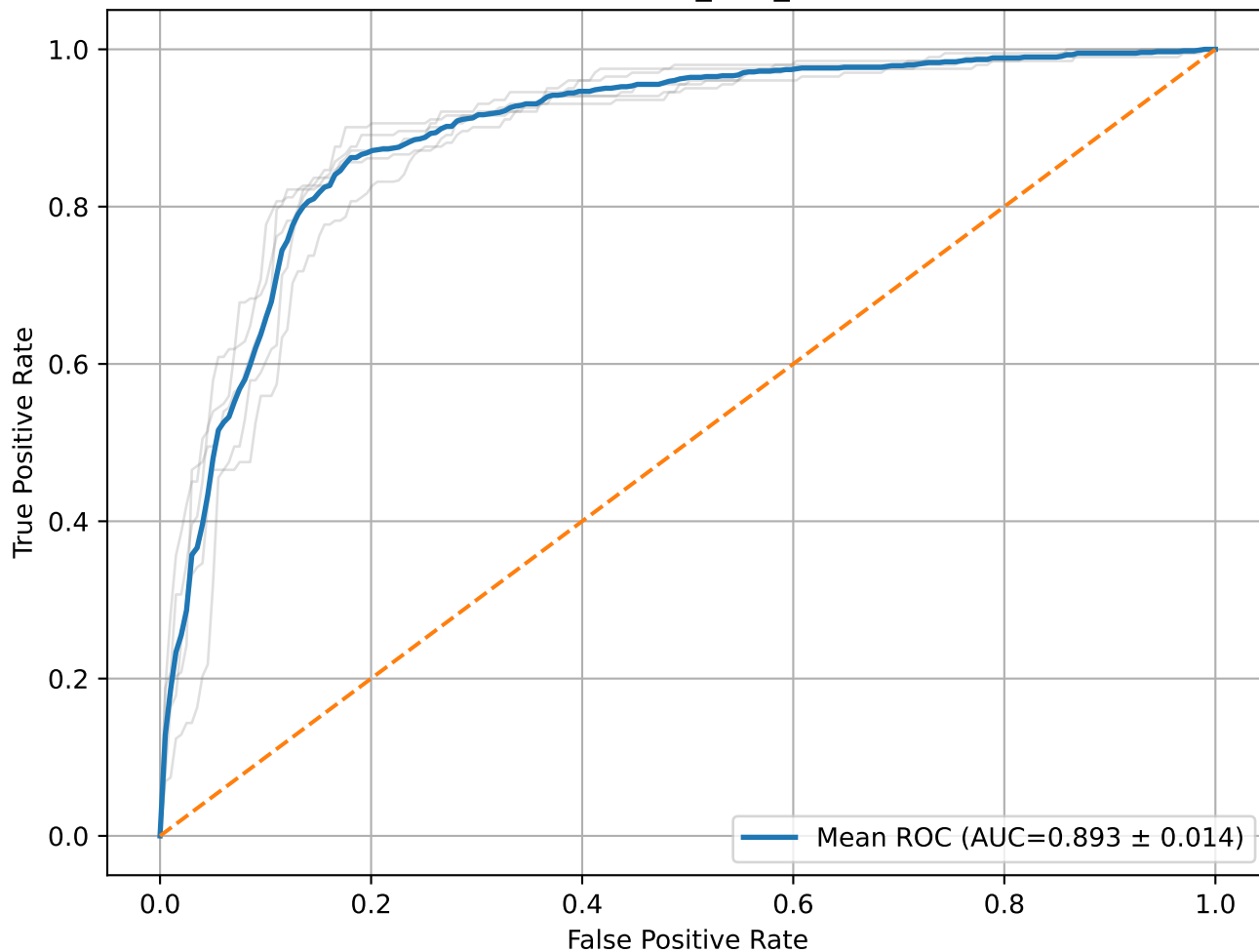
Confusion Matrix - PSO_SVM (last fold)



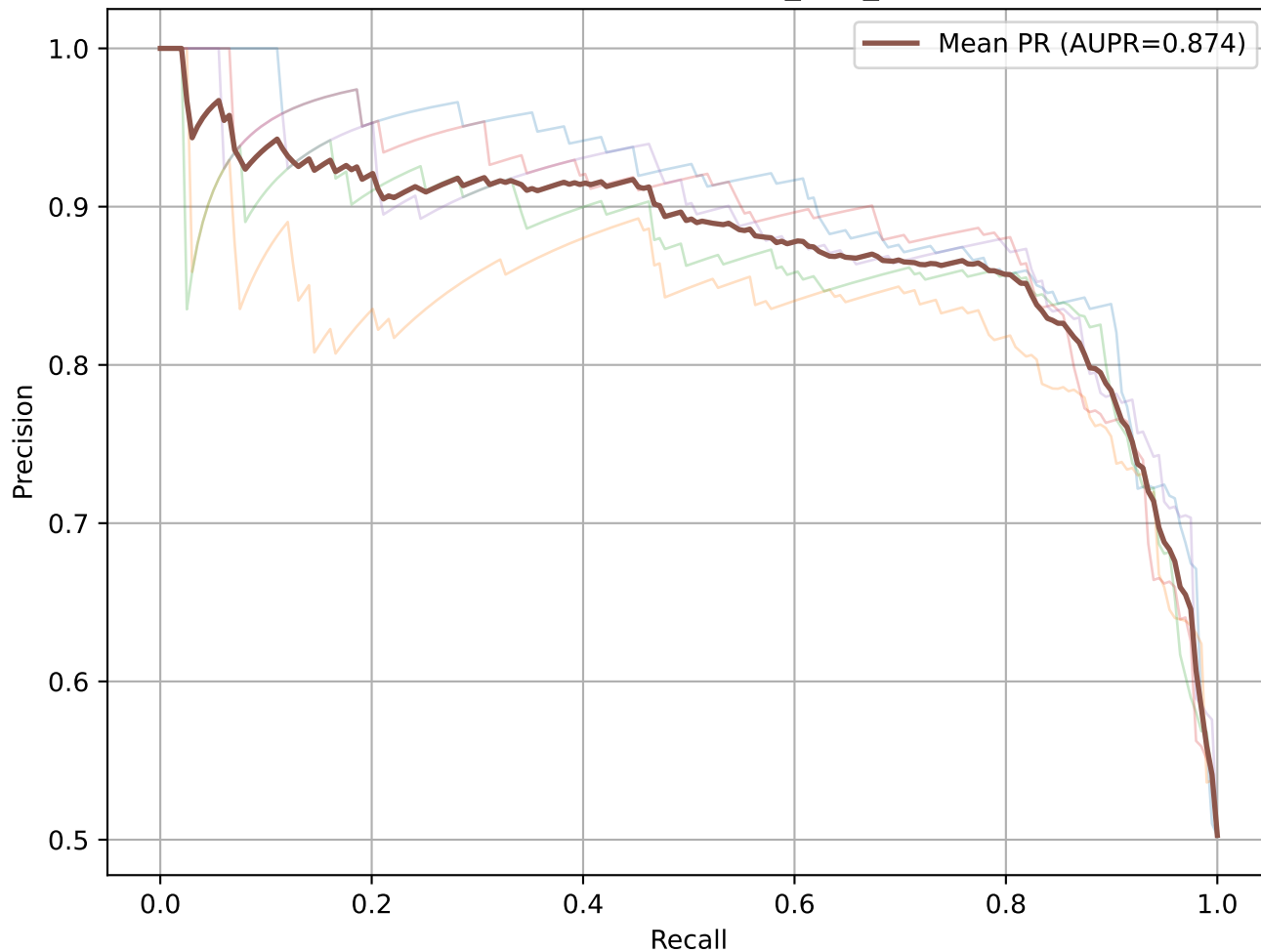
ClassificationReport - PSO_SVM (last fold)

	precision	recall	f1-score	support
0	0.82	0.80	0.81	199
1	0.81	0.83	0.82	202
accuracy			0.82	401
macro avg	0.82	0.82	0.82	401
weighted avg	0.82	0.82	0.82	401

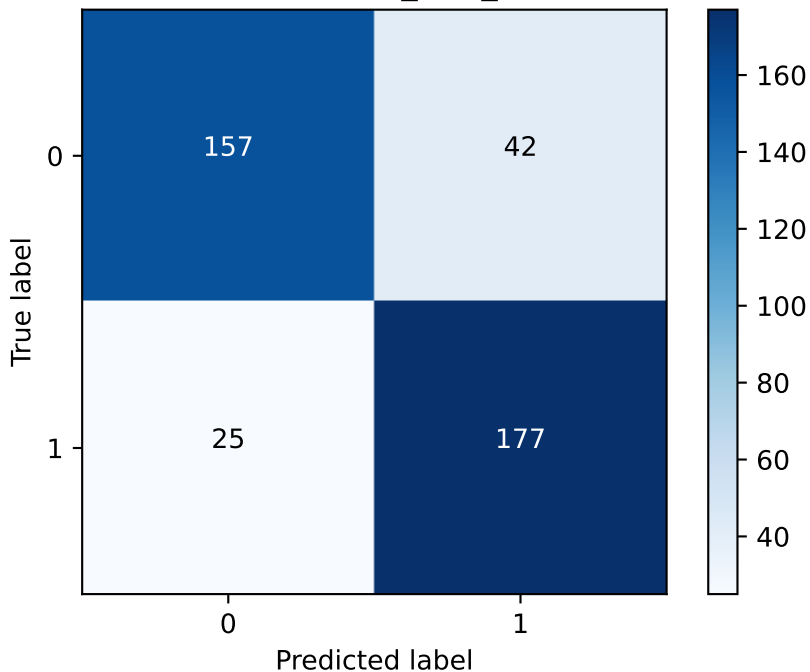
ROC - BAYES_OPT_SVMS



Precision-Recall - BAYES_OPT_SVMS



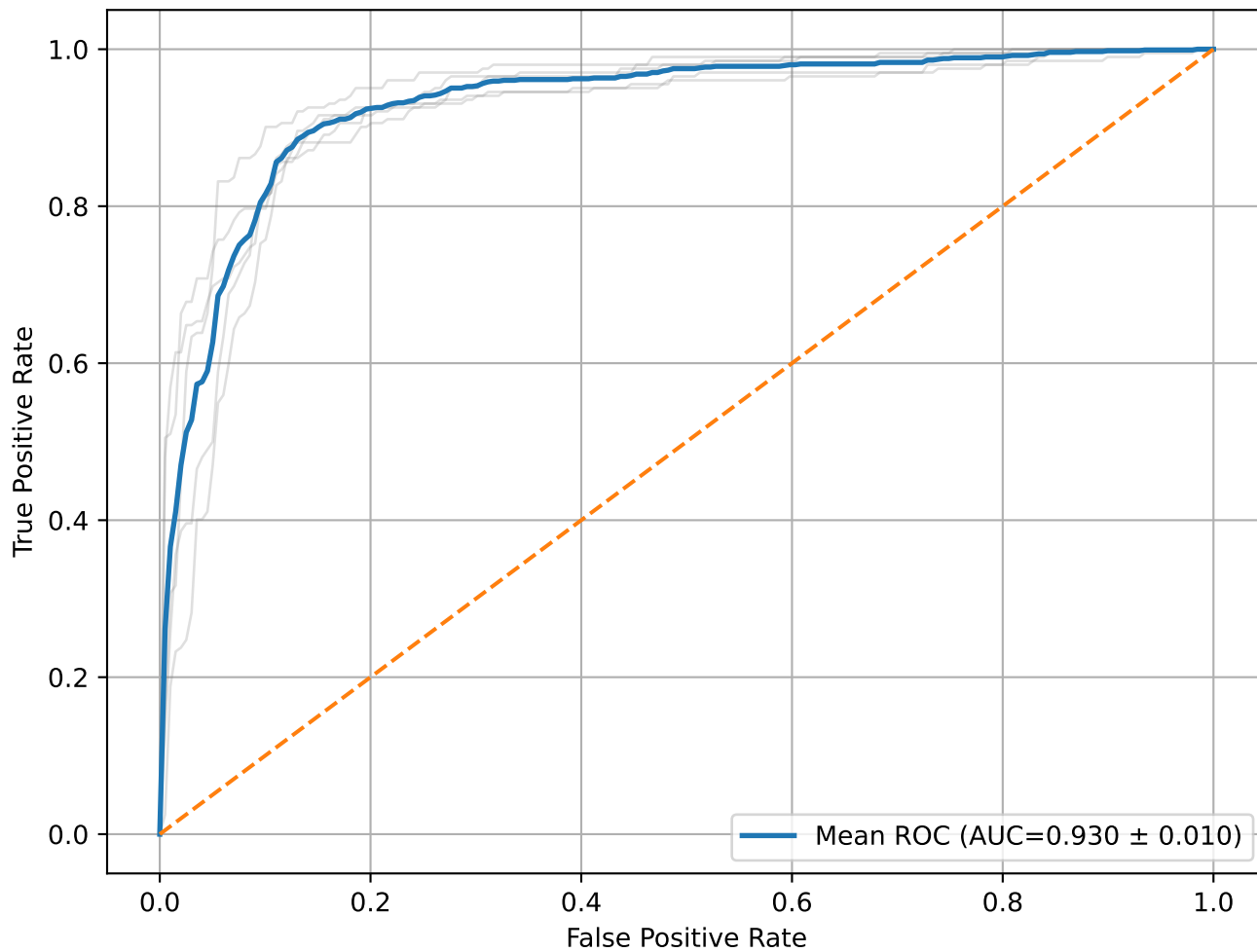
Confusion Matrix - BAYES_OPT_SVMS (last fold)



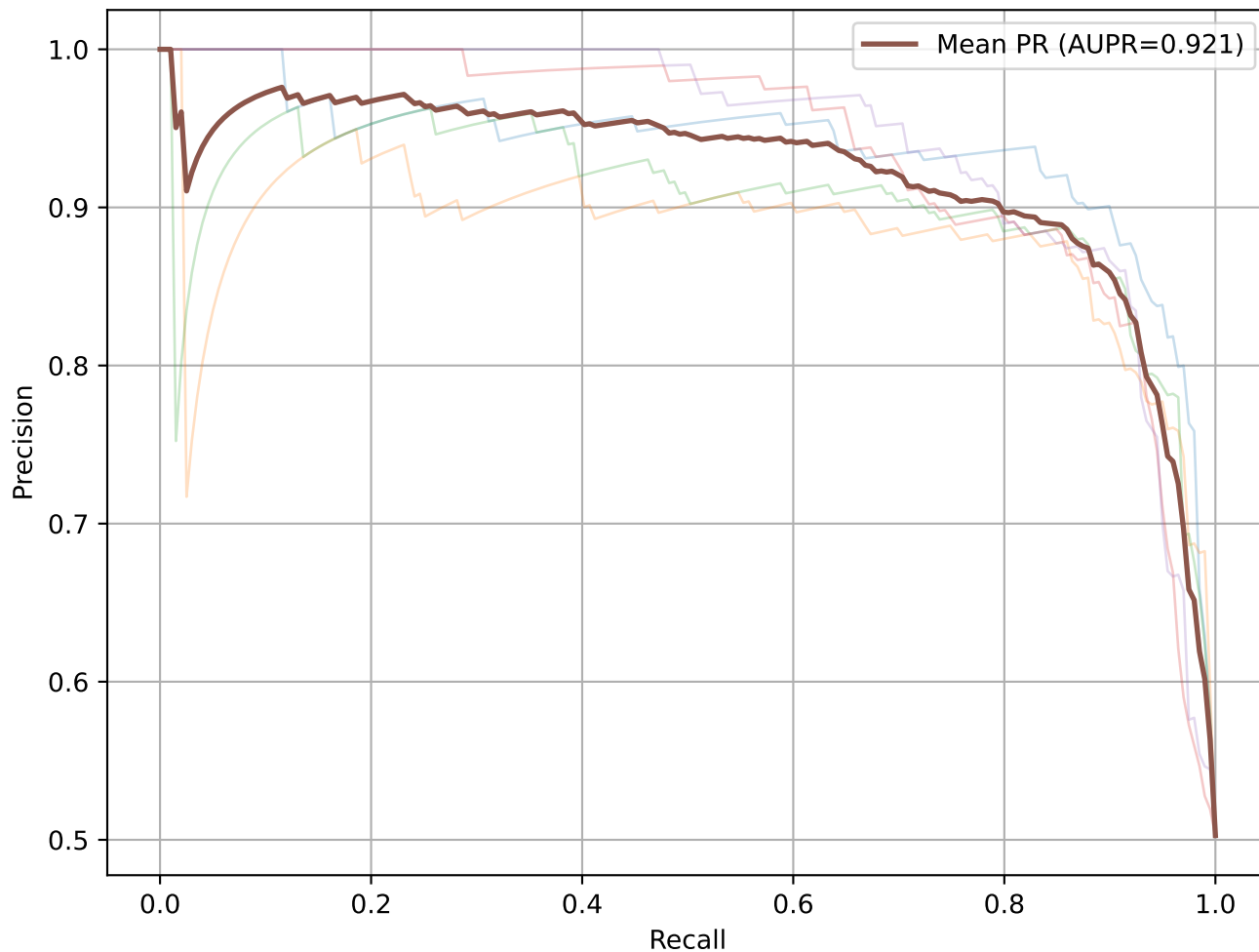
ClassificationReport - BAYES_OPT_SVMS (last fold)

	precision	recall	f1-score	support
0	0.86	0.79	0.82	199
1	0.81	0.88	0.84	202
accuracy			0.83	401
macro avg	0.84	0.83	0.83	401
weighted avg	0.84	0.83	0.83	401

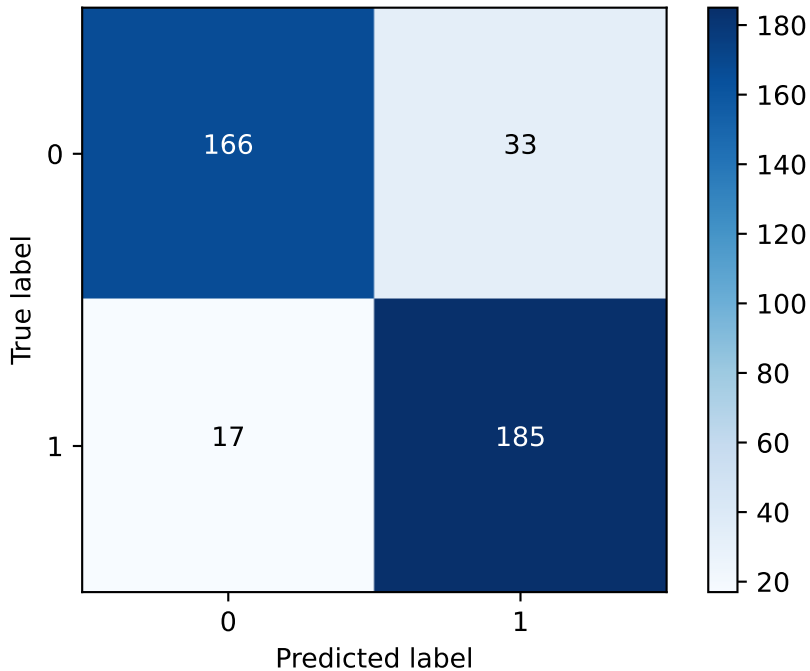
ROC - GBDT



Precision-Recall - GBDT



Confusion Matrix - GBDT (last fold)



ClassificationReport - GBDT (last fold)

	precision	recall	f1-score	support
0	0.91	0.83	0.87	199
1	0.85	0.92	0.88	202
accuracy			0.88	401
macro avg	0.88	0.88	0.88	401
weighted avg	0.88	0.88	0.88	401

Model Comparison Table (sorted by mean ROC AUC)

model	mean_roc_auc	std_roc_auc	mean_aupr	mean_accuracy	mean_f1
lgb	0.935739	0.008866	0.932707	0.877055	0.878947
ga_xgb	0.930812	0.009819	0.923023	0.875561	0.878174
gbdt	0.928620	0.010272	0.920911	0.864615	0.870645
rf	0.928341	0.007825	0.928149	0.884521	0.885928
voting	0.928196	0.011079	0.922966	0.877554	0.880481
xgb	0.927915	0.010375	0.916748	0.871586	0.874229
stacking	0.922600	0.013554	0.911655	0.872072	0.877430
bagging	0.905393	0.013415	0.897492	0.832750	0.832393
logistic	0.896484	0.017069	0.880111	0.836240	0.836877
bayes_opt_svms	0.890974	0.013629	0.873929	0.832255	0.835623
pca_svm	0.886401	0.039376	0.879325	0.828284	0.825088
dnn	0.882221	0.013639	0.876745	0.814326	0.808690
svm	0.878466	0.032649	0.869642	0.817831	0.813923
pso_svm	0.862545	0.033927	0.864588	0.782994	0.773558

Combined ROC - All Models

